

Bridging the Gap Between Anchor-based and Anchor-free Detection via Adaptive Training Sample Selection

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Abstract

*Object detection has been dominated by anchor-based detectors for several years. Recently, anchor-free detectors have become popular due to the proposal of FPN and Focal Loss. In this paper, we first point out that the essential difference between anchor-based and anchor-free detection is actually **how to define positive and negative training samples**, which leads to the performance gap between them. If they adopt the same definition of positive and negative samples during training, there is no obvious difference in the final performance, no matter regressing from a box or a point. This shows that how to select positive and negative training samples is important for current object detectors. Then, we propose an Adaptive Training Sample Selection (ATSS) to automatically select positive and negative samples according to statistical characteristics of object. It significantly improves the performance of anchor-based and anchor-free detectors and bridges the gap between them. Finally, we discuss the necessity of tiling multiple anchors per location on the image to detect objects. Extensive experiments conducted on MS COCO support our aforementioned analysis and conclusions. With the newly introduced ATSS, we improve state-of-the-art detectors by a large margin to 50.7% AP without introducing any overhead. The code is available at <https://github.com/sfzhang15/ATSS>.*

1. Introduction

Object detection is a long-standing topic in the field of computer vision, aiming to detect objects of predefined categories. Accurate object detection would have far reaching impact on various applications including image recognition and video surveillance. In recent years, with the development of convolutional neural network (CNN), object detection has been dominated by anchor-based detectors, which can be generally divided into one-stage methods [36, 33]

and two-stage methods [47, 9]. Both of them first tile a large number of preset anchors on the image, then predict the category and refine the coordinates of these anchors by one or several times, finally output these refined anchors as detection results. Because two-stage methods refine anchors several times more than one-stage methods, the former one has more accurate results while the latter one has higher computational efficiency. State-of-the-art results on common detection benchmarks are still held by anchor-based detectors.

Recent academic attention has been geared toward anchor-free detectors due to the emergence of FPN [32] and Focal Loss [33]. Anchor-free detectors directly find objects without preset anchors in two different ways. One way is to first locate several pre-defined or self-learned keypoints and then bound the spatial extent of objects. We call this type of anchor-free detectors as keypoint-based methods [26, 71]. Another way is to use the center point or region of objects to define positives and then predict the four distances from positives to the object boundary. We call this kind of anchor-free detectors as center-based methods [56, 23]. These anchor-free detectors are able to eliminate those hyperparameters related to anchors and have achieved similar performance with anchor-based detectors, making them more potential in terms of generalization ability.

Among these two types of anchor-free detectors, keypoint-based methods follow the standard keypoint estimation pipeline that is different from anchor-based detectors. However, center-based detectors are similar to anchor-based detectors, which treat points as preset samples instead of anchor boxes. Take the one-stage anchor-based detector RetinaNet [33] and the center-based anchor-free detector FCOS [56] as an example, there are three main differences between them: (1) The number of anchors tiled per location. RetinaNet tiles several anchor boxes per location, while FCOS tiles one anchor point¹ per location. (2) The definition of positive and negative samples. RetinaNet resorts to the Intersection over Union (IoU) for positives and

¹A point in FCOS is equal to the center of an anchor box in RetinaNet, thus we call it as the anchor point. A pair of anchor point and box is associated to the same location of feature map to be classified and regressed.

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基于锚点与无锚点检测之间的差距通过自适应训练样本选择来弥补

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摘要

*Object detection has been dominated by anchor-based detectors for several years. Recently, anchor-free detectors have become popular due to the proposal of FPN and Focal Loss. In this paper, we first point out that the essential difference between anchor-based and anchor-free detection is actually **how to define positive and negative training samples**, which leads to the performance gap between them. If they adopt the same definition of positive and negative samples during training, there is no obvious difference in the final performance, no matter regressing from a box or a point. This shows that how to select positive and negative training samples is important for current object detectors. Then, we propose an Adaptive Training Sample Selection (ATSS) to automatically select positive and negative samples according to statistical characteristics of object. It significantly improves the performance of anchor-based and anchor-free detectors and bridges the gap between them. Finally, we discuss the necessity of tiling multiple anchors per location on the image to detect objects. Extensive experiments conducted on MS COCO support our aforementioned analysis and conclusions. With the newly introduced ATSS, we improve state-of-the-art detectors by a large margin to 50.7% AP without introducing any overhead. The code is available at <https://github.com/sfzhang15/ATSS>.*

1. 引言

目标检测是计算机视觉领域一个长期存在的主题，旨在检测预定义类别的物体。准确的目标检测将对图像识别和视频监控等多种应用产生深远影响。近年来，随着卷积神经网络（CNN）的发展，目标检测领域主要由基于锚框的检测器主导，这些方法通常可分为单阶段方法[36, 33]

以及两阶段方法[47, 9]。两者都首先在图像上平铺大量预设锚框，随后通过一次或多次迭代预测这些锚框的类别并优化其坐标，最终将这些优化后的锚框输出为检测结果。由于两阶段方法优化锚框的次数多于单阶段方法，前者具有更精确的结果，而后者则具备更高的计算效率。目前常见检测基准的最先进成果仍由基于锚框的检测器保持。

近期，由于特征金字塔网络（FPN）[32]和焦点损失（Focal Loss）[33]的出现，学术界的注意力逐渐转向了无锚框检测器。无锚框检测器通过两种不同的方式直接定位目标，而无需预设锚框。一种方式是先定位若干预定义或自学习的关键点，然后框定目标的空间范围。我们将这类无锚框检测器称为基于关键点的方法[26, 71]。另一种方式则是利用目标的中心点或中心区域来定义正样本，进而预测正样本到目标边界的四个距离。我们将这类无锚框检测器称为基于中心点的方法[56, 23]。这些无锚框检测器能够消除与锚框相关的超参数，并已取得与基于锚框的检测器相近的性能，从而在泛化能力方面展现出更大的潜力。

在这两种无锚检测器中，基于关键点的方法遵循标准的关键点估计流程，这与基于锚框的检测器不同。然而，基于中心点的检测器则与基于锚框的检测器相似，它们将点视为预设样本而非锚框。以单阶段基于锚框的检测器RetinaNet [33]和基于中心点的无锚检测器FCOS [56]为例，它们之间存在三个主要差异：(1) 每个位置平铺的锚点数量。RetinaNet在每个位置平铺多个锚框，而FCOS在每个位置仅平铺一个锚点¹。(2) 正负样本的定义。RetinaNet依赖交并比（IoU）来确定正样本和

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negatives, while FCOS utilizes spatial and scale constraints to select samples. (3) The regression starting status. RetinaNet regresses the object bounding box from the preset anchor box, while FCOS locates the object from the anchor point. As reported in [56], the anchor-free FCOS achieves much better performance than the anchor-based RetinaNet, it is worth studying which of these three differences are essential factors for the performance gap.

In this paper, we investigate the differences between anchor-based and anchor-free methods in a fair way by strictly ruling out all the implementation inconsistencies between them. It can be concluded from experiment results that the essential difference between these two kind of methods is the definition of positive and negative training samples, which results in the performance gap between them. If they select the same positive and negative samples during training, there is no obvious gap in the final performance, no matter regressing from a box or a point. Therefore, how to select positive and negative training samples deserves further study. Inspired by that, we propose a new Adaptive Training Sample Selection (ATSS) to automatically select positive and negative samples based on object characteristics. It bridges the gap between anchor-based and anchor-free detectors. Besides, through a series of experiments, a conclusion can be drawn that tiling multiple anchors per location on the image to detect objects is not necessary. Extensive experiments on the MS COCO [34] dataset support our analysis and conclusions. State-of-the-art AP 50.7% is achieved by applying the newly introduced ATSS without introducing any overhead. The main contributions of this work can be summarized as:

- Indicating the essential difference between anchor-based and anchor-free detectors is actually how to define positive and negative training samples.
- Proposing an adaptive training sample selection to automatically select positive and negative training samples according to statistical characteristics of object.
- Demonstrating that tiling multiple anchors per location on the image to detect objects is a useless operation.
- Achieving state-of-the-art performance on MS COCO without introducing any additional overhead.

2. Related Work

Current CNN-based object detection consists of anchor-based and anchor-free detectors. The former one can be divided into two-stage and one-stage methods, while the latter one falls into keypoint-based and center-based methods.

2.1. Anchor-based Detector

Two-stage method. The emergence of Faster R-CNN [47] establishes the dominant position of two-stage anchor-based detectors. Faster R-CNN consists of a separate region

proposal network (RPN) and a region-wise prediction network (R-CNN) [14, 13] to detect objects. After that, lots of algorithms are proposed to improve its performance, including architecture redesign and reform [4, 9, 5, 28, 30], context and attention mechanism [2, 51, 38, 7, 44], multi-scale training and testing [54, 41], training strategy and loss function [40, 52, 61, 17], feature fusion and enhancement [25, 32], better proposal and balance [55, 43]. Nowadays, state-of-the-art results are still held by two-stage anchor-based methods on standard detection benchmarks.

One-stage method. With the advent of SSD [36], one-stage anchor-based detectors have attracted much attention because of their high computational efficiency. SSD spreads out anchor boxes on multi-scale layers within a ConvNet to directly predict object category and anchor box offsets. Thereafter, plenty of works are presented to boost its performance in different aspects, such as fusing context information from different layers [24, 12, 69], training from scratch [50, 73], introducing new loss function [33, 6], anchor refinement and matching [66, 67], architecture redesign [21, 22], feature enrichment and alignment [35, 68, 60, 42, 29]. At present, one-stage anchor-based methods can achieve very close performance with two-stage anchor-based methods at a faster inference speed.

2.2. Anchor-free Detector

Keypoint-based method. This type of anchor-free method first locates several pre-defined or self-learned keypoints, and then generates bounding boxes to detect objects. CornerNet [26] detects an object bounding box as a pair of keypoints (top-left corner and bottom-right corner) and CornerNet-Lite [27] introduces CornerNet-Saccade and CornerNet-Squeeze to improve its speed. The second stage of Grid R-CNN [39] locates objects via predicting grid points with the position sensitive merits of FCN and then determining the bounding box guided by the grid. ExtremeNet [71] detects four extreme points (top-most, left-most, bottom-most, right-most) and one center point to generate the object bounding box. Zhu *et al.* [70] use keypoint estimation to find center point of objects and regress to all other properties including size, 3D location, orientation and pose. CenterNet [11] extends CornetNet as a triplet rather than a pair of keypoints to improve both precision and recall. RepPoints [65] represents objects as a set of sample points and learns to arrange themselves in a manner that bounds the spatial extent of an object and indicates semantically significant local areas.

Center-based method. This kind of anchor-free method regards the center (*e.g.*, the center point or part) of object as foreground to define positives, and then predicts the distances from positives to the four sides of the object bounding box for detection. YOLO [45] divides the image into an $S \times S$ grid, and the grid cell that contains the center of

负样本，而FCOS利用空间和尺度约束来选择样本。(3) 回归起始状态。RetinaNet从预设的锚框回归目标边界框，而FCOS则从锚点定位目标。如[56]所述，无锚框的FCOS实现了比基于锚框的RetinaNet更好的性能，值得研究这三种差异中哪些是导致性能差距的关键因素。

本文通过严格排除基于锚点与无锚点方法之间的所有实现不一致性，以公平方式探究两者的本质差异。实验结果表明，两类方法的根本区别在于正负训练样本的定义方式，这直接导致了性能差距的产生。若在训练过程中采用相同的正负样本选择策略，则无论从边界框还是单点进行回归，最终性能均无显著差异。因此，如何选择正负训练样本值得深入研究。受此启发，我们提出一种基于目标特征自适应选择正负样本的新方法（ATSS），该方法弥合了基于锚点与无锚点检测器之间的鸿沟。此外，通过系列实验可得出结论：在图像每个位置平铺多个锚点并非目标检测的必要操作。在MS COCO数据集上的大量实验验证了我们的分析与结论。应用新提出的ATSS方法无需引入额外计算开销，即实现了50.7%的先进AP指标。本研究的主要贡献可归纳为：

- 基于锚框和无锚框检测器的本质区别实际上在于如何定义正负训练样本。
- 提出一种自适应训练样本选择方法，根据目标的统计特性自动选择正负训练样本。
- 证明在图像的每个位置上平铺多个锚点来检测物体是一种无用的操作。
- 在MS COCO上实现最先进的性能，且不引入任何额外开销。

2. 相关工作

当前基于CNN的目标检测方法主要分为基于锚框和无锚框检测器两大类。前者可进一步划分为两阶段和单阶段方法，而后者则分为基于关键点和基于中心点的方法。

2.1. 基于锚点的检测器

两阶段方法。Faster R-CNN [47] 的出现确立了基于锚点的两阶段检测器的主导地位。Faster R-CNN 由一个独立的区域

区域提议网络（RPN）和区域预测网络（R-CNN）[14, 13]被用于检测物体。此后，大量算法被提出以提升其性能，包括架构的重新设计与改进[4, 9, 5, 28, 30]、上下文与注意力机制[2, 51, 38, 7, 44]、多尺度训练与测试[54, 41]、训练策略与损失函数[40, 52, 61, 17]、特征融合与增强[25, 32]，以及更优的提议与平衡方法[55, 43]。目前，在标准检测基准上，最先进的结果仍由基于锚点的两阶段方法保持。

单阶段方法。随着SSD[36]的出现，单阶段基于锚点的检测器因其高计算效率而备受关注。SSD在卷积网络的多尺度层上散布锚框，直接预测物体类别和锚框偏移。此后，大量研究从不同方面提升其性能，例如融合不同层的上下文信息[24, 12, 69]、从头训练[50, 73]、引入新的损失函数[33, 6]、锚点优化与匹配[66, 67]、架构重新设计[21, 22]、特征增强与对齐[35, 68, 60, 42, 29]。目前，单阶段基于锚点的方法能以更快的推理速度，达到与两阶段基于锚点方法非常接近的性能。

2.2. 无锚框检测器

基于关键点的方法。这类无锚框方法首先定位若干预定义或自学习的关键点，随后生成边界框以检测目标。CornerNet [26] 将目标边界框检测为一对关键点（左上角与右下角），而CornerNet-Lite [27] 通过引入CornerNet-Saccade与CornerNet-Squeeze来提升速度。Grid R-CNN [39] 的第二阶段利用全卷积网络的位置敏感特性预测网格点来定位目标，再通过网格引导确定边界框。ExtremeNet [71] 检测四个极值点（最顶部、最左侧、最底部、最右侧）和一个中心点以生成目标边界框。Zhu *et al.* [70] 采用关键点估计寻找目标中心点，并回归其他所有属性包括尺寸、3D位置、方向与姿态。CenterNet [11] 将CornerNet扩展为三元关键点而非二元关键点，从而同时提升精确率与召回率。RepPoints [65] 将目标表示为一组采样点，并学习以包围目标空间范围且指示语义显著局部区域的方式排列这些点。

基于中心点的方法。这类无锚框方法将目标的中心（*e.g.*，即中心点或部分）视为前景以定义正样本，随后通过预测正样本到目标边界框四边的距离进行检测。YOLO [45] 将图像划分为 $S \times S$ 的网格，并将包含目标中心的网格单元

an object is responsible for detecting this object. DenseBox [20] uses a filled circle located in the center of the object to define positives and then predicts the four distances from positives to the bound of the object bounding box for location. GA-RPN [59] defines the pixels in the center region of the object as positives to predict the location, width and height of object proposals for Faster R-CNN. FSAF [72] attaches an anchor-free branch with online feature selection to RetinaNet. The newly added branch defines the center region of the object as positives to locate it via predicting four distances to its bounds. FCOS [56] regards all the locations inside the object bounding box as positives with four distances and a novel centerness score to detect objects. CSP [37] only defines the center point of the object box as positives to detect pedestrians with fixed aspect ratio. FoveaBox [23] regards the locations in the middle part of object as positives with four distances to perform detection.

3. Difference Analysis of Anchor-based and Anchor-free Detection

Without loss of generality, the representative anchor-based RetinaNet [33] and anchor-free FCOS [56] are adopted to dissect their differences. In this section, we focus on the last two differences: the positive/negative sample definition and the regression starting status. The remaining one difference: the number of anchors tiled per location, will be discussed in subsequent section. Thus, we just tile one square anchor per location for RetinaNet, which is quite similar to FCOS. In the remaining part, we first introduce the experiment settings, then rule out all the implementation inconsistencies, finally point out the essential difference between anchor-based and anchor-free detectors.

3.1. Experiment Setting

Dataset. All experiments are conducted on the challenging MS COCO [34] dataset that includes 80 object classes. Following the common practice [33, 56], all 115K images in the `trainval35k` split is used for training, and all 5K images in the `minival` split is used as validation for analysis study. We also submit our main results to the evaluation server for the final performance on the `test-dev` split.

Training Detail. We use the ImageNet [49] pretrained ResNet-50 [16] with 5-level feature pyramid structure as the backbone. The newly added layers are initialized in the same way as in [33]. For RetinaNet, each layer in the 5-level feature pyramid is associated with one square anchor with $8S$ scale, where S is the total stride size. During training, we resize the input images to keep their shorter side being 800 and their longer side less or equal to 1,333. The whole network is trained using the Stochastic Gradient Descent (SGD) algorithm for 90K iterations with 0.9 momentum, 0.0001 weight decay and 16 batch size. We set the initial learning rate as 0.01 and decay it by 0.1 at iteration 60K

Table 1: Analysis of implementation inconsistencies between RetinaNet and FCOS on MS COCO `minival` set. “#A=1” means there is one square anchor box per location.

Inconsistency	FCOS	RetinaNet (#A=1)					
GroupNorm	✓	✓	✓	✓	✓	✓	✓
GIoU Loss	✓		✓	✓	✓	✓	✓
In GT Box	✓			✓	✓	✓	✓
Centerness	✓				✓	✓	✓
Scalar	✓						✓
AP (%)	37.8	32.5	33.4	34.9	35.3	36.8	37.0

and 80K, respectively. Unless otherwise stated, the aforementioned training details are used in the experiments.

Inference Detail. During the inference phase, we resize the input image in the same way as in the training phase, and then forward it through the whole network to output the predicted bounding boxes with a predicted class. After that, we use the preset score 0.05 to filter out plenty of background bounding boxes, and then output the top 1000 detections per feature pyramid. Finally, the Non-Maximum Suppression (NMS) is applied with the IoU threshold 0.6 per class to generate final top 100 confident detections per image.

3.2. Inconsistency Removal

We mark the anchor-based detector RetinaNet with only one square anchor box per location as **RetinaNet (#A=1)**, which is almost the same as the anchor-free detector FCOS. However, as reported in [56], FCOS outperforms RetinaNet (#A=1) by a large margin in AP performance on the MS COCO `minival` subset, *i.e.*, 37.1% vs. 32.5%. Furthermore, some new improvements have been made for FCOS including moving centerness to regression branch, using GIoU loss function and normalizing regression targets by corresponding strides. These improvements boost the AP performance of FCOS from 37.1% to 37.8%², making the gap even bigger. However, part of the AP gap between the anchor-based detector (32.5%) and the anchor-free detector (37.8%) results from some universal improvements that are proposed or used in FCOS, such as adding GroupNorm [62] in heads, using the GIoU [48] regression loss function, limiting positive samples in the ground-truth box [56], introducing the centerness branch [56] and adding a trainable scalar [56] for each level feature pyramid. These improvements can also be applied to anchor-based detectors, therefore they are not the essential differences between anchor-based and anchor-free methods. We apply them to RetinaNet (#A=1) one by one so as to rule out these implementation inconsistencies. As listed in Table 1, these irrelevant

²This 37.8% AP result does not include the center sample improvement, which is our contribution that has been merged into FCOS and will be introduced in Sec. 4.2.

一个物体负责检测该物体。DenseBox [20] 使用位于物体中心的实心圆来定义正样本，然后预测正样本到物体边界框四边的距离以进行定位。GA-RPN [59] 将物体中心区域的像素定义为正样本，用于预测Faster R-CNN中物体提议的位置、宽度和高度。FSAF [72] 在RetinaNet上附加了一个带有在线特征选择的无锚分支。这个新增的分支将物体的中心区域定义为正样本，通过预测到其边界的四个距离来定位物体。FCOS [56] 将物体边界框内的所有位置都视为正样本，并使用四个距离和一个新颖的中心度分数来检测物体。CSP [37] 仅将物体框的中心点定义为正样本，以固定宽高比检测行人。FoveaBox [23] 将物体中间部分的位置视为正样本，通过四个距离进行检测。

3. 基于锚点与无锚点检测的差异分析

在不失一般性的前提下，我们采用具有代表性的基于锚点的RetinaNet [33] 和无锚点的FCOS [56] 来剖析二者的差异。本节中，我们聚焦于最后两项差异：正负样本的定义以及回归起始状态。剩余的一项差异——每个位置平铺的锚点数量——将在后续章节中讨论。因此，我们仅为RetinaNet的每个位置平铺一个方形锚点，这与FCOS非常相似。在接下来的部分中，我们将首先介绍实验设置，然后排除所有实现上的一致性，最后指出基于锚点与无锚点检测器之间的本质区别。

3.1. 实验设置

数据集。所有实验均在具有挑战性的MS COCO [34]数据集上进行，该数据集包含80个物体类别。遵循常规做法[33, 56]，我们使用trainval35k分割中的所有115K张图像进行训练，并使用minival分割中的所有5K张图像作为分析研究的验证集。我们还将主要结果提交至评估服务器，以获取在test-dev分割上的最终性能。

训练细节。我们使用ImageNet [49]预训练的ResNet-50 [16]作为主干网络，并采用5层特征金字塔结构。新增层的初始化方式与[33]相同。对于RetinaNet，5层特征金字塔中的每一层都关联一个尺度为 $8S$ 的方形锚框，其中 S 为总步长。训练过程中，我们将输入图像调整为短边保持800像素，长边不超过1,333像素。整个网络采用随机梯度下降（SGD）算法进行训练，共进行90K次迭代，动量为0.9，权重衰减为0.0001，批量大小为16。初始学习率设为0.01，并在第60K次迭代时衰减为0.1倍。

表1：在MS COCO minival集上对RetinaNet与FCOS实现不一致性的分析。“ $\#A\{v\}1$ ”表示每个位置有一个方形锚框。

Inconsistency	FCOS	RetinaNet ($\#A=1$)						
GroupNorm	✓	✓	✓	✓	✓	✓	✓	✓
GIoU Loss	✓		✓	✓	✓	✓	✓	✓
In GT Box	✓			✓	✓	✓	✓	✓
Centerness	✓				✓	✓	✓	✓
Scalar	✓							✓
AP (%)	37.8	32.5	33.4	34.9	35.3	36.8	37.0	

分别为80K。除非另有说明，实验中均采用上述训练细节。推理细节。在推理阶段，我们以与训练阶段相同的方式调整输入图像的尺寸，然后将其通过整个网络前向传播，输出带有预测类别的边界框。之后，我们使用预设分数0.05过滤掉大量背景边界框，然后为每个特征金字塔输出前1000个检测结果。最后，对每个类别应用交并比阈值为0.6的非极大值抑制，以生成每张图像最终的前100个高置信度检测结果。

3.2. 不一致性消除

我们将每个位置仅使用一个正方形锚框的基于锚点的检测器RetinaNet标记为RetinaNet ($\#A=1$)，其与无锚检测器FCOS几乎相同。然而，如[56]所述，FCOS在MS COCO minival子集上的AP性能大幅超越RetinaNet ($\#A=1$)，*i.e.*, 37.1% vs. 32.5%。此外，FCOS还进行了一些新的改进，包括将中心度移至回归分支、使用GIoU损失函数以及通过相应步长对回归目标进行归一化。这些改进将FCOS的AP性能从37.1%提升至37.8%²，使得差距进一步扩大。然而，基于锚点的检测器（32.5%）与无锚检测器（37.8%）之间的部分AP差距源于FCOS提出或采用的一些通用改进，例如在头部添加GroupNorm [62]、使用GIoU [48]回归损失函数、限制正样本在真实框内[56]、引入中心度分支[56]以及为每个层级特征金字塔添加可训练标量[56]。这些改进同样可应用于基于锚点的检测器，因此它们并非基于锚点与无锚方法之间的本质差异。我们逐一将这些改进应用于RetinaNet ($\#A=1$)，以排除这些实现不一致性。如表1所列，这些无关的

²This 37.8% AP result does not include the center sample improvement, which is our contribution that has been merged into FCOS and will be introduced in Sec. 4.2.

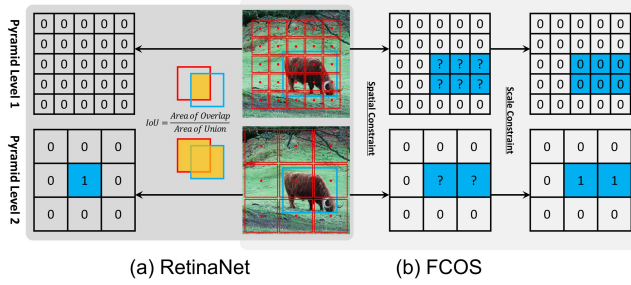


Figure 1: Definition of positives (1) and negatives (0). Blue box, red box and red point are ground-truth, anchor box and anchor point. (a) RetinaNet uses IoU to select positives (1) in spatial and scale dimension simultaneously. (b) FCOS first finds candidate positives (1) in spatial dimension, then selects final positives (1) in scale dimension.

differences improve the anchor-based RetinaNet to 37.0%, which still has a gap of 0.8% to the anchor-free FCOS. By now, after removing all the irrelevant differences, we can explore the essential differences between anchor-based and anchor-free detectors in a quite fair way.

3.3. Essential Difference

After applying those universal improvements, these are only two differences between the anchor-based RetinaNet (#A=1) and the anchor-free FCOS. One is about the classification sub-task in detection, *i.e.*, the way to define positive and negative samples. Another one is about the regression sub-task, *i.e.*, the regression starting from an anchor box or an anchor point.

Classification. As shown in Figure 1(a), RetinaNet utilizes IoU to divide the anchor boxes from different pyramid levels into positives and negatives. It first labels the best anchor box of each object and the anchor boxes with $\text{IoU} > \theta_p$ as positives, then regards the anchor boxes with $\text{IoU} < \theta_n$ as negatives, finally other anchor boxes are ignored during training. As shown in Figure 1(b), FCOS uses spatial and scale constraints to divide the anchor points from different pyramid levels. It first considers the anchor points within the ground-truth box as candidate positive samples, then selects the final positive samples from candidates based on the scale range defined for each pyramid level³, finally those unselected anchor points are negative samples.

As shown in Figure 1, FCOS first uses the spatial constraint to find candidate positives in the spatial dimension, then uses the scale constraint to select final positives in the scale dimension. In contrast, RetinaNet utilizes IoU to directly select the final positives in the spatial and scale dimension simultaneously. These two different sample selec-

³There are several preset hyperparameters in FCOS to define the scale range for five pyramid levels: $[m2, m3]$ for P3, $[m3, m4]$ for P4, $[m4, m5]$ for P5, $[m5, m6]$ for P6 and $[m6, m7]$ for P7.

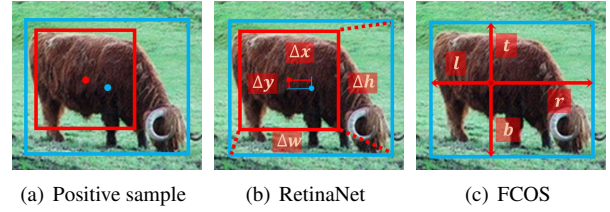


Figure 2: (a) Blue point and box are the center and bound of object, red point and box are the center and bound of anchor. (b) RetinaNet regresses from anchor box with four offsets. (c) FCOS regresses from anchor point with four distances.

Table 2: Analysis of differences (%) between RetinaNet and FCOS on the MS COCO minival set.

	Regression	Box	Point
Classification			
Intersection over Union		37.0	36.9
Spatial and Scale Constraint		37.8	37.8

tion strategies produce different positive and negative samples. As listed in the first column of Table 2 for RetinaNet (#A=1), using the spatial and scale constraint strategy instead of the IoU strategy improves the AP performance from 37.0% to 37.8%. As for FCOS, if it uses the IoU strategy to select positive samples, the AP performance decreases from 37.8% to 36.9% as listed in the second column of Table 2. These results demonstrate that the definition of positive and negative samples is an essential difference between anchor-based and anchor-free detectors.

Regression. After positive and negative samples are determined, the location of object is regressed from positive samples as shown in Figure 2(a). RetinaNet regresses from the anchor box with four offsets between the anchor box and the object box as shown in Figure 2(b), while FCOS regresses from the anchor point with four distances to the bound of object as shown in Figure 2(c). It means that for a positive sample, the regression starting status of RetinaNet is a box while FCOS is a point. However, as shown in the first and second rows of Table 2, when RetinaNet and FCOS adopt the same sample selection strategy to have consistent positive/negative samples, there is no obvious difference in final performance, no matter regressing starting from a point or a box, *i.e.*, 37.0% vs. 36.9% and 37.8% vs. 37.8%. These results indicate that the regression starting status is an irrelevant difference rather than an essential difference.

Conclusion. According to these experiments conducted in a fair way, we indicate that the essential difference between one-stage anchor-based detectors and center-based anchor-free detectors is actually how to define positive and negative training samples, which is important for current object detection and deserves further study.

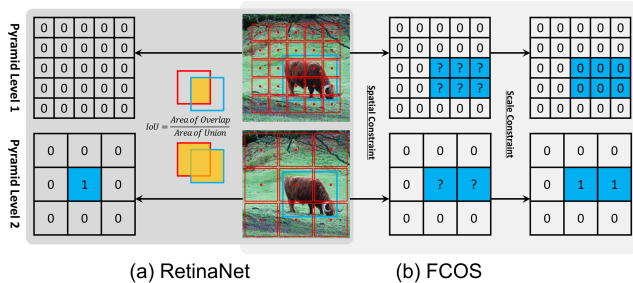


图1: 正样本 (i) 与负样本 (o) 的定义。蓝色框、红色框与红点分别表示真实框、锚框与锚点。(a) RetinaNet 使用交并比在空间与尺度维度上同时筛选正样本 (i)。(b) FCOS 首先在空间维度确定候选正样本 (i)，随后在尺度维度筛选最终正样本 (i)。

差异将基于锚点的RetinaNet提升至37.0%，但与无锚点的FCOS仍存在0.8%的差距。至此，在排除所有无关差异后，我们得以在相当公平的条件探究基于锚点与无锚点检测器之间的本质差异。

3.3. 本质区别

在应用了这些通用改进之后，基于锚点的RetinaNet (#A=1) 与无锚点的FCOS之间仅存在两点差异。一是关于检测中的分类子任务，*i.e.*，即定义正负样本的方式；另一是关于回归子任务，*i.e.*，即从锚框或锚点开始的回归方法。

分类。如图1(a)所示，RetinaNet利用IoU将来自不同金字塔层级的锚框划分为正样本和负样本。它首先将每个目标的最佳锚框及IoU大于等于 θ_p 的锚框标记为正样本，随后将IoU小于 θ_n 的锚框视为负样本，最后其余锚框在训练过程中被忽略。如图1(b)所示，FCOS采用空间和尺度约束来划分不同金字塔层级的锚点。它首先将真实框内的锚点视为候选正样本，然后根据为每个金字塔层级定义的尺度范围³从候选样本中筛选出最终正样本，最后未被选中的锚点即为负样本。

如图1所示，FCOS首先利用空间约束在空间维度寻找候选正样本，随后通过尺度约束在尺度维度筛选最终正样本。相比之下，RetinaNet直接利用IoU在空间与尺度维度上同步选择最终正样本。这两种不同的样本选择

³There are several preset hyperparameters in FCOS to define the scale range for five pyramid levels: $[m_2, m_3]$ for P3, $[m_3, m_4]$ for P4, $[m_4, m_5]$ for P5, $[m_5, m_6]$ for P6 and $[m_6, m_7]$ for P7.

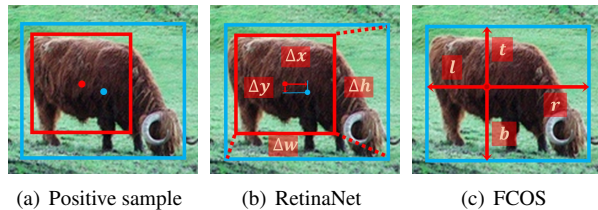


图2: (a) 蓝点和框是目标的中心与边界，红点和框是锚点的中心与边界。(b) RetinaNet通过四个偏移量从锚框回归。(c) FCOS通过四个距离从锚点回归。

表2: RetinaNet与FCOS在MS COCO minival集上的差异分析 (%)。

	Regression	Box	Point
Classification			
Intersection over Union		37.0	36.9
Spatial and Scale Constraint		37.8	37.8

不同的正负样本采样策略会产生不同的正负样本。如表2第一列RetinaNet (#A=1) 所示，使用空间和尺度约束策略代替IoU策略将AP性能从37.0%提升至37.8%。对于FCOS，若采用IoU策略选择正样本，其AP性能会从37.8%下降至36.9%（见表2第二列）。这些结果表明，正负样本的定义是基于锚点与无锚点检测器之间的本质区别。

回归。在正负样本确定后，从正样本回归目标位置如图2(a)所示。RetinaNet从锚框回归四个锚框与目标框之间的偏移量如图2(b)，而FCOS从锚点回归四个到目标边界的距离如图2(c)。这意味着对于正样本，RetinaNet的回归起始状态是一个框，而FCOS是一个点。然而，如表2前两行所示，当RetinaNet和FCOS采用相同的样本选择策略以获得一致的正负样本时，无论从点还是框开始回归，最终性能没有明显差异，*i.e.*、37.0% vs. 36.9% 和 37.8% vs. 37.8%。这些结果表明，回归起始状态是一个无关紧要的差异，而非本质区别。

结论。根据这些以公平方式进行的实验，我们指出，基于锚框的单阶段检测器与基于中心的无锚框检测器之间的本质区别实际上在于如何定义正负训练样本，这对当前的目标检测至关重要，值得进一步研究。

4. Adaptive Training Sample Selection

When training an object detector, we first need to define positive and negative samples for classification, and then use positive samples for regression. According to the previous analysis, the former one is crucial and the anchor-free detector FCOS improves this step. It introduces a new way to define positives and negatives, which achieves better performance than the traditional IoU-based strategy. Inspired by this, we delve into the most basic issue in object detection: *how to define positive and negative training samples*, and propose an Adaptive Training Sample Selection (ATSS). Compared with these traditional strategies, our method almost has no hyperparameters and is robust to different settings.

4.1. Description

Previous sample selection strategies have some sensitive hyperparameters, such as IoU thresholds in anchor-based detectors and scale ranges in anchor-free detectors. After these hyperparameters are set, all ground-truth boxes must select their positive samples based on the fixed rules, which are suitable for most objects, but some outer objects will be neglected. Thus, different settings of these hyperparameters will have very different results.

To this end, we propose the ATSS method that automatically divides positive and negative samples according to statistical characteristics of object almost without any hyperparameter. Algorithm 1 describes how the proposed method works for an input image. For each ground-truth box g on the image, we first find out its candidate positive samples. As described in Line 3 to 6, on each pyramid level, we select k anchor boxes whose center are closest to the center of g based on L2 distance. Supposing there are $\mathcal{L}$ feature pyramid levels, the ground-truth box g will have $k \times \mathcal{L}$ candidate positive samples. After that, we compute the IoU between these candidates and the ground-truth g as $\mathcal{D}_g$ in Line 7, whose mean and standard deviation are computed as m_g and v_g in Line 8 and Line 9. With these statistics, the IoU threshold for this ground-truth g is obtained as $t_g = m_g + v_g$ in Line 10. Finally, we select these candidates whose IoU are greater than or equal to the threshold t_g as final positive samples in Line 11 to 15. Notably, we also limit the positive samples' center to the ground-truth box as shown in Line 12. Besides, if an anchor box is assigned to multiple ground-truth boxes, the one with the highest IoU will be selected. The rest are negative samples. Some motivations behind our method are explained as follows.

Selecting candidates based on the center distance between anchor box and object. For RetinaNet, the IoU is larger when the center of anchor box is closer to the center of object. For FCOS, the closer anchor point to the center of object will produce higher-quality detections. Thus, the closer anchor to the center of object is the better candidate.

Algorithm 1 Adaptive Training Sample Selection (ATSS)

Input:

- $\mathcal{G}$ is a set of ground-truth boxes on the image
- $\mathcal{L}$ is the number of feature pyramid levels
- $\mathcal{A}_i$ is a set of anchor boxes from the i_{th} pyramid levels
- $\mathcal{A}$ is a set of all anchor boxes
- k is a quite robust hyperparameter with a default value of 9

Output:

- $\mathcal{P}$ is a set of positive samples
- $\mathcal{N}$ is a set of negative samples

```

1: for each ground-truth  $g \in \mathcal{G}$  do
2:   build an empty set for candidate positive samples of the
     ground-truth  $g$ :  $\mathcal{C}_g \leftarrow \emptyset$ ;
3:   for each level  $i \in [1, \mathcal{L}]$  do
4:      $\mathcal{S}_i \leftarrow$  select  $k$  anchors from  $\mathcal{A}_i$  whose center are closest
       to the center of ground-truth  $g$  based on L2 distance;
5:      $\mathcal{C}_g = \mathcal{C}_g \cup \mathcal{S}_i$ ;
6:   end for
7:   compute IoU between  $\mathcal{C}_g$  and  $g$ :  $\mathcal{D}_g = \text{IoU}(\mathcal{C}_g, g)$ ;
8:   compute mean of  $\mathcal{D}_g$ :  $m_g = \text{Mean}(\mathcal{D}_g)$ ;
9:   compute standard deviation of  $\mathcal{D}_g$ :  $v_g = \text{Std}(\mathcal{D}_g)$ ;
10:  compute IoU threshold for ground-truth  $g$ :  $t_g = m_g + v_g$ ;
11:  for each candidate  $c \in \mathcal{C}_g$  do
12:    if  $\text{IoU}(c, g) \geq t_g$  and center of  $c$  in  $g$  then
13:       $\mathcal{P} = \mathcal{P} \cup c$ ;
14:    end if
15:  end for
16: end for
17:  $\mathcal{N} = \mathcal{A} - \mathcal{P}$ ;
18: return  $\mathcal{P}, \mathcal{N}$ ;

```

Using the sum of mean and standard deviation as the IoU threshold. The IoU mean m_g of an object is a measure of the suitability of the preset anchors for this object. A high m_g as shown in Figure 3(a) indicates it has high-quality candidates and the IoU threshold is supposed to be high. A low m_g as shown in Figure 3(b) indicates that most of its candidates are low-quality and the IoU threshold should be low. Besides, the IoU standard deviation v_g of an object is a measure of which layers are suitable to detect this object. A high v_g as shown in Figure 3(a) means there is a pyramid level specifically suitable for this object, adding v_g to m_g obtains a high threshold to select positives only from that level. A low v_g as shown in Figure 3(b) means that there are several pyramid levels suitable for this object, adding v_g to m_g obtains a low threshold to select appropriate positives from these levels. Using the sum of mean m_g and standard deviation v_g as the IoU threshold t_g can adaptively select enough positives for each object from appropriate pyramid levels in accordance of statistical characteristics of object.

Limiting the positive samples' center to object. The anchor with a center outside object is a poor candidate and will be predicted by the features outside the object, which is not conducive to training and should be excluded.

4. 自适应训练样本选择

在训练目标检测器时，我们首先需要为分类任务定义正负样本，随后利用正样本进行回归。根据先前的分析，前者至关重要，而无锚框检测器FCOS改进了这一步骤。它引入了一种定义正负样本的新方法，其性能优于传统的基于IoU的策略。受此启发，我们深入探究目标检测中最基本的问题：

how to define positive and negative training samples，并提出了一种自适应训练样本选择方法（ATSS）。与传统策略相比，我们的方法几乎无需超参数，且对不同设置具有鲁棒性。

4.1. 描述

以往的样本选择策略存在一些敏感的超参数，例如基于锚点的检测器中的IoU阈值和无锚点检测器中的尺度范围。这些超参数一旦设定，所有真实框都必须依据固定规则选择其正样本，这些规则适用于大多数目标，但部分边缘目标会被忽略。因此，这些超参数的不同设置会导致差异显著的结果。

为此，我们提出了ATSS方法，该方法根据目标的统计特征自动划分正负样本，几乎不需要任何超参数。算法1描述了所提方法对输入图像的处理流程。对于图像上的每个真实框 g ，我们首先找出其候选正样本。如第3至6行所述，在每个金字塔层级上，我们基于L2距离选取 k 个中心点最接近 g 中心的锚框。假设共有 $\mathcal{L}$ 个特征金字塔层级，则真实框 g 将拥有 $k \times \mathcal{L}$ 个候选正样本。随后，我们计算这些候选样本与真实框 g 之间的IoU值，记为 $\mathcal{D}_g$ （第7行），其均值和标准差分别在第8行和第9行计算为 m_g 和 v_g 。基于这些统计量，该真实框 g 的IoU阈值在第10行确定为 $t_g = m_g + v_g$ 。最后，我们在第11至15行选取IoU大于等于阈值 t_g 的候选样本作为最终正样本。值得注意的是，如第12行所示，我们还将正样本的中心点限制在真实框内。此外，若一个锚框被分配给多个真实框，则选择IoU最高的匹配。其余样本作为负样本。我们方法背后的部分设计动机如下所述。

根据锚框与目标中心距离选择候选框。对于RetinaNet，当锚框中心越接近目标中心时，IoU值越大。对于FCOS，越接近目标中心的锚点会产生更高质量的检测结果。因此，越接近目标中心的锚框是更优的候选框。

Algorithm 1 Adaptive Training Sample Selection (ATSS)

Input:

$\mathcal{G}$ is a set of ground-truth boxes on the image
 $\mathcal{L}$ is the number of feature pyramid levels
 $\mathcal{A}_i$ is a set of anchor boxes from the i_{th} pyramid levels
 $\mathcal{A}$ is a set of all anchor boxes
 k is a quite robust hyperparameter with a default value of 9

Output:

$\mathcal{P}$ is a set of positive samples
 $\mathcal{N}$ is a set of negative samples

```

1: for each ground-truth  $g \in \mathcal{G}$  do
2:   build an empty set for candidate positive samples of the
     ground-truth  $g$ :  $\mathcal{C}_g \leftarrow \emptyset$ ;
3:   for each level  $i \in [1, \mathcal{L}]$  do
4:      $\mathcal{S}_i \leftarrow$  select  $k$  anchors from  $\mathcal{A}_i$  whose center are closest
        to the center of ground-truth  $g$  based on L2 distance;
5:      $\mathcal{C}_g = \mathcal{C}_g \cup \mathcal{S}_i$ ;
6:   end for
7:   compute IoU between  $\mathcal{C}_g$  and  $g$ :  $\mathcal{D}_g = \text{IoU}(\mathcal{C}_g, g)$ ;
8:   compute mean of  $\mathcal{D}_g$ :  $m_g = \text{Mean}(\mathcal{D}_g)$ ;
9:   compute standard deviation of  $\mathcal{D}_g$ :  $v_g = \text{Std}(\mathcal{D}_g)$ ;
10:  compute IoU threshold for ground-truth  $g$ :  $t_g = m_g + v_g$ ;
11:  for each candidate  $c \in \mathcal{C}_g$  do
12:    if  $\text{IoU}(c, g) \geq t_g$  and center of  $c$  in  $g$  then
13:       $\mathcal{P} = \mathcal{P} \cup c$ ;
14:    end if
15:  end for
16: end for
17:  $\mathcal{N} = \mathcal{A} - \mathcal{P}$ ;
18: return  $\mathcal{P}, \mathcal{N}$ ;

```

使用均值与标准差之和作为IoU阈值。物体的IoU均值 m_g 是衡量预设锚框对该物体适用程度的指标。如图3(a)所示，较高的 m_g 意味着存在高质量候选框，此时应设置较高的IoU阈值。如图3(b)所示，较低的 m_g 表明其多数候选框质量较低，此时IoU阈值应相应降低。此外，物体的IoU标准差 v_g 用于衡量哪些特征层级适合检测该物体。如图3(a)所示，较高的 v_g 表示存在特别适合该物体的金字塔层级，将 v_g 与 m_g 相加可获得较高阈值，从而仅从该层级筛选正样本。如图3(b)所示，较低的 v_g 意味着有多个金字塔层级均适合检测该物体，将 v_g 与 m_g 相加可获得较低阈值，以便从这些层级中选取合适的正样本。采用均值 m_g 与标准差 v_g 之和作为IoU阈值 t_g ，能够依据物体的统计特性，自适应地从合适的金字塔层级为每个物体选取足量正样本。

将正样本中心限制在物体内部。中心位于物体之外的锚点是一个较差的候选，将由物体外部的特征进行预测，这不利于训练，应当被排除。

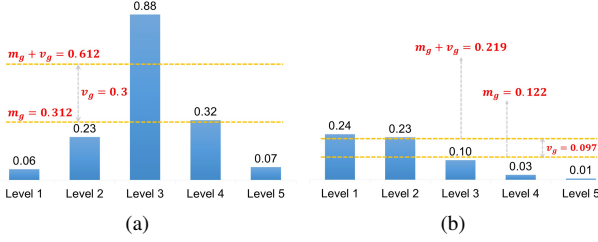


Figure 3: Illustration of ATSS. Each level has one candidate with its IoU. (a) A ground-truth with a high m_g and a high v_g . (b) A ground-truth with a low m_g and a low v_g .

Maintaining fairness between different objects. According to the statistical theory⁴, about 16% of samples are in the confidence interval $[m_g + v_g, 1]$ in theory. Although the IoU of candidates is not a standard normal distribution, the statistical results show that each object has about $0.2 * k\mathcal{L}$ positive samples, which is invariant to its scale, aspect ratio and location. In contrast, strategies of RetinaNet and FCOS tend to have much more positive samples for larger objects, leading to unfairness between different objects.

Keeping almost hyperparameter-free. Our method only has one hyperparameter k . Subsequent experiments prove that it is quite insensitive to the variations of k and the proposed ATSS can be considered almost hyperparameter-free.

4.2. Verification

Anchor-based RetinaNet. To verify the effectiveness of our adaptive training sample selection for anchor-based detectors, we use it to replace the traditional strategy in the improved RetinaNet (#A=1). As shown in Table 3, it consistently boosts the performance by 2.3% on AP, 2.4% on AP₅₀, 2.9% for AP₇₅, 2.9% for AP_S, 2.1% for AP_M and 2.7% for AP_L. These improvements are mainly due to the adaptive selection of positive samples for each ground-truth based on its statistical characteristics. Since our method only redefines positive and negative samples without incurring any additional overhead, these improvements can be considered cost-free.

Anchor-free FCOS. The proposed method can also be applied to the anchor-free FCOS in two different versions: the lite and full version. For the lite version, we apply some ideas of the proposed ATSS to FCOS, *i.e.*, replacing its way to select candidate positives with the way in our method. FCOS considers anchor points in the object box as candidates, which results in plenty of low-quality positives. In contrast, our method selects top $k = 9$ candidates per pyramid level for each ground-truth. The lite version of our method has been merged to the official code of FCOS as the center sampling, which improves FCOS from 37.8% to

Table 3: Verification of the proposed method (%) on the MS COCO minival set. ATSS and center sampling are the full version and the lite version of our proposed method.

Method	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
RetinaNet (#A=1)	37.0	55.1	39.9	21.4	41.2	48.6
RetinaNet (#A=1) + ATSS	39.3	57.5	42.8	24.3	43.3	51.3
FCOS	37.8	55.6	40.7	22.1	41.8	48.8
FCOS + Center sampling	38.6	57.4	41.4	22.3	42.5	49.8
FCOS + ATSS	39.2	57.3	42.4	22.7	43.1	51.5

38.6% on AP as listed in Table 3. However, the hyperparameters of scale ranges still exist in the lite version.

For the full version, we let the anchor point in FCOS become the anchor box with 8S scale to define positive and negative samples, then still regress these positive samples to objects from the anchor point like FCOS. As shown in Table 3, it significantly increases the performance by 1.4% for AP, by 1.7% for AP₅₀, by 1.7% for AP₇₅, by 0.6% for AP_S, by 1.3% for AP_M and by 2.7% for AP_L. Notably, these two versions have the same candidates selected in the spatial dimension, but different ways to select final positives from candidates along the scale dimension. As listed in the last two rows of Table 3, the full version (ATSS) outperforms the lite version (center sampling) across different metrics by a large margin. These results indicate that the adaptive way in our method is better than the fixed way in FCOS to select positives from candidates along the scale dimension.

4.3. Analysis

Training an object detector with the proposed adaptive training sample selection only involves one hyperparameter k and one related setting of anchor boxes. This subsection analyzes them one after another.

Hyperparameter k . We conduct several experiments to study the robustness of the hyperparameter k , which is used to select the candidate positive samples from each pyramid level. As shown in Table 4, different values of k in [3, 5, 7, 9, 11, 13, 15, 17, 19] are used to train the detector. We observe that the proposed method is quite insensitive to the variations of k from 7 to 17. Too large k (*e.g.*, 19) will result in too many low-quality candidates that slightly decreases the performance. Too small k (*e.g.*, 3) causes a noticeable drop in accuracy, because too few candidate positive samples will cause statistical instability. Overall, the only hyperparameter k is quite robust and the proposed ATSS can be nearly regarded as hyperparameter-free.

Table 4: Analysis of different values of hyperparameter k on the MS COCO minival set.

k	3	5	7	9	11	13	15	17	19
AP (%)	38.0	38.8	39.1	39.3	39.1	39.0	39.1	39.2	38.9

⁴<http://dwz1.cc/sNigLI2>

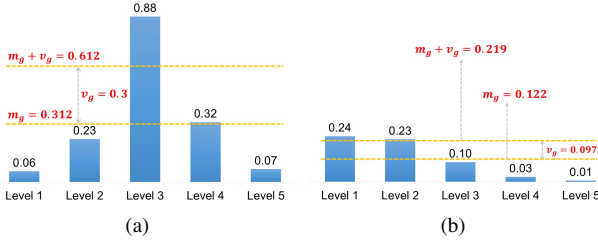


图3: ATSS示意图。每个层级都有一个候选框及其IoU值。(a) 一个具有高 m_g 和高 v_g 的真实框。(b) 一个具有低 m_g 和低 v_g 的真实框。

在不同物体之间保持公平性。根据统计理论⁴, 理论上约有16%的样本处于置信区间 $[m_g + v_g, 1]$ 内。尽管候选框的交并比并非标准正态分布, 但统计结果显示每个物体约有 $0.2 * k\mathcal{L}$ 个正样本, 且该数量不随物体尺度、长宽比或位置变化。相比之下, RetinaNet和FCOS的策略往往为较大物体分配更多正样本, 导致不同物体间存在不公平性。

保持几乎无需超参数调整。我们的方法仅有一个超参数 k 。后续实验证明, 该方法对 k 的变化相当不敏感, 因此所提出的ATSS可视为几乎无需超参数调整的方法。

4.2. 验证

基于锚点的RetinaNet。为验证我们提出的自适应训练样本选择方法在基于锚点的检测器上的有效性, 我们将其应用于改进版RetinaNet (#A=1)中以替代传统策略。如表3所示, 该方法在AP指标上稳定提升2.3%, AP₅₀提升2.4%, AP₇₅提升2.9%, AP_S提升2.9%, AP_M提升2.1%, AP_L提升2.7%。这些提升主要源于我们根据每个真实标注框的统计特征自适应选择正样本。由于该方法仅重新定义正负样本而未引入额外计算开销, 这些性能提升可视为零成本获得。

无锚框FCOS。所提出的方法同样可以应用于无锚框FCOS, 并分为两个不同版本: 轻量版和完整版。对于轻量版, 我们将ATSS的部分思路应用于FCOS, *i.e.*, 用我们方法中的候选正样本选择方式替换其原有方式。FCOS将目标框内的锚点视为候选, 这会导致大量低质量正样本。相比之下, 我们的方法为每个真实标注在金字塔层级上选取前 $k=9$ 个候选样本。该方法的轻量版已作为中心采样功能并入FCOS官方代码库, 将FCOS的精度从37.8%提升至

表3: 在MS COCO minival集上对所提方法的验证(%)。ATSS和中心采样分别是我们所提方法的完整版和轻量版。

Method	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
RetinaNet (#A=1)	37.0	55.1	39.9	21.4	41.2	48.6
RetinaNet (#A=1) + ATSS	39.3	57.5	42.8	24.3	43.3	51.3
FCOS	37.8	55.6	40.7	22.1	41.8	48.8
FCOS + Center sampling	38.6	57.4	41.4	22.3	42.5	49.8
FCOS + ATSS	39.2	57.3	42.4	22.7	43.1	51.5

如表3所列, AP上为38.6%。然而, 精简版中仍存在尺度范围的超参数。

在完整版本中, 我们将FCOS中的锚点扩展为具有8 S尺度的锚框来定义正负样本, 随后仍像FCOS一样从锚点出发对这些正样本进行目标回归。如表3所示, 该改进显著提升了各项性能指标: AP提升1.4%, AP₅₀提升1.7%, AP₇₅提升1.7%, AP_S提升0.6%, AP_M提升1.3%, AP_L提升2.7%。值得注意的是, 这两个版本在空间维度上选择的候选区域相同, 但在尺度维度上从候选区域筛选最终正样本的方式不同。如表3最后两行所示, 完整版本(ATSS)在所有评估指标上均大幅优于轻量版本(中心采样)。这些结果表明, 在尺度维度上从候选区域选择正样本时, 我们方法中的自适应策略优于FCOS中的固定策略。

4.3. 分析

使用提出的自适应训练样本选择方法训练目标检测器仅涉及一个超参数 k 和一个锚框相关设置。本节将逐一分析它们。

超参数 k 。我们进行了若干实验来研究超参数 k 的鲁棒性, 该参数用于从每个金字塔层级选择候选正样本。如表4所示, 我们使用[3, 5, 7, 9, 11, 13, 15, 17, 19]中的不同 k 值来训练检测器。我们观察到, 所提出的方法对 k 在7到17之间的变化相当不敏感。过大的 k (*e.g.*, 即19), 会导致过多低质量候选样本, 从而略微降低性能。过小的 k (*e.g.*, 即3), 则会导致准确率显著下降, 因为候选正样本过少会引起统计不稳定性。总体而言, 唯一的超参数 k 具有相当强的鲁棒性, 所提出的ATSS方法几乎可视为无超参数方法。

表4: 在MS COCO minival集上分析超参数 k 不同取值的表现。

k	3	5	7	9	11	13	15	17	19
AP (%)	38.0	38.8	39.1	39.3	39.1	39.0	39.1	39.2	38.9

⁴<http://dwz1.cc/sNIgLI2>

Table 5: Analysis (%) of different anchor scales with fixed aspect ratio 1 : 1 on the MS COCO minival set.

Scale	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
5	39.0	57.9	41.9	23.2	42.8	50.5
6	39.2	57.6	42.5	23.5	42.8	51.1
7	39.3	57.6	42.4	22.9	43.2	51.3
8	39.3	57.5	42.8	24.3	43.3	51.3
9	38.9	56.5	42.0	22.9	42.4	50.3

Table 6: Analysis (%) of different anchor aspect ratios with fixed scale 8 S on the MS COCO minival set.

Aspect Ratio	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
1:4	39.1	57.2	42.3	23.1	43.1	51.4
1:2	39.0	56.9	42.5	23.3	43.5	50.6
1:1	39.3	57.5	42.8	24.3	43.3	51.3
2:1	39.3	57.4	42.3	22.8	43.4	51.0
4:1	39.1	56.9	42.6	22.9	42.9	50.7

Anchor Size. The introduced method resorts to the anchor boxes to define positives and we also study the effect of the anchor size. In the previous experiments, one square anchor with 8 S (S indicates the total stride size of the pyramid level) is tiled per location. As shown in Table 5, we conduct some experiments with different scales of the square anchor in [5, 6, 7, 8, 9] and the performances are quite stable. Besides, several experiments with different aspect ratios of the 8 S anchor box are performed as shown in Table 6. The performances are also insensitive to this variation. These results indicate that the proposed method is robust to different anchor settings.

4.4. Comparison

We compare our final models on the MS COCO test-dev subset in Table 8 with other state-of-the-art object detectors. Following previous works [33, 56], the multi-scale training strategy is adopted for these experiments, *i.e.*, randomly selecting a scale between 640 to 800 to resize the shorter side of images during training. Besides, we double the total number of iterations to 180 K and the learning rate reduction points to 120 K and 160 K correspondingly. Other settings are consistent with those mentioned before.

As shown in Table 8, our method with ResNet-101 achieves 43.6% AP without any bells and whistles, which is better than all the methods with the same backbone including Cascade R-CNN [5] (42.8% AP), C-Mask RCNN [7] (42.0% AP), RetinaNet [33] (39.1% AP) and RefineDet [66] (36.4% AP). We can further improve the AP accuracy of the proposed method to 45.1% and 45.6% by using larger backbone networks ResNeXt-32x8d-101 and ResNeXt-64x4d-101 [63], respectively. The 45.6% AP result surpasses all the anchor-free and anchor-based detectors except only 0.1% lower than SNIP [54] (45.7% AP), which introduces the improved multi-scale training

Table 7: Results (%) with different multiple anchors per location on the MS COCO minival set.

Method	#sc	#ar	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
RetinaNet (#A=9)	3	3	36.3	55.2	38.8	19.8	39.8	48.8
+Imprs.	3	3	38.4	56.2	41.6	22.2	42.4	50.1
+Imprs.+ATSS	3	3	39.2	57.6	42.7	23.8	42.8	50.9
+Imprs.+ATSS	3	1	39.3	57.7	42.6	23.8	43.5	51.2
+Imprs.+ATSS	1	3	39.2	57.1	42.5	23.2	43.1	50.3
+Imprs.+ATSS	1	1	39.3	57.5	42.8	24.3	43.3	51.3

and testing strategy. Since our method is about the definition of positive and negative samples, it is compatible and complementary to most of current technologies. We further use the Deformable Convolutional Networks (DCN) [10] to the ResNet and ResNeXt backbones as well as the last layer of detector towers. DCN consistently improves the AP performances to 46.3% for ResNet-101, 47.7% for ResNeXt-32x8d-101 and 47.7% for ResNeXt-64x4d-101, respectively. The best result 47.7% is achieved with single-model and single-scale testing, outperforming all the previous detectors by a large margin. Finally, with the multi-scale testing strategy, our best model achieves 50.7% AP.

4.5. Discussion

Previous experiments are based on RetinaNet with only one anchor per location. There is still a difference between anchor-based and anchor-free detectors that is not explored: the number of anchors tiled per location. Actually, the original RetinaNet tiles 9 anchors (3 scales $\times$ 3 aspect ratios) per location (marked as **RetinaNet (#A=9)**) that achieves 36.3% AP as listed in the first row of Table 7. In addition, those universal improvements in Table 1 can also be used to RetinaNet (#A=9), boosting the AP performance from 36.3% to 38.4%. Without using the proposed ATSS, the improved RetinaNet (#A=9) has better performance than RetinaNet (#A=1), *i.e.*, 38.4% in Table 7 vs. 37.0% in Table 1. These results indicate that under the traditional IoU-based sample selection strategy, tiling more anchor box per location is effective.

However, after using our proposed method, the opposite conclusion will be drawn. To be specific, the proposed ATSS also improves RetinaNet (#A=9) by 0.8% on AP, 1.4% on AP₅₀ and 1.1% on AP₇₅, achieving similar performances to RetinaNet (#A=1) as listed in the third and sixth rows of Table 7. Besides, when we change the number of anchor scales or aspect ratios from 3 to 1, the results are almost unchanged as listed in the fourth and fifth rows of Table 7. In other words, as long as the positive samples are selected appropriately, no matter how many anchors are tiled at each location, the results are the same. We argue that tiling multiple anchors per location is a useless operation under our proposed method and it needs further study to discover its right role.

表5: 在MS COCO minival集上固定宽高比1:1时不同锚框尺度的分析 (%)。

Scale	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
5	39.0	57.9	41.9	23.2	42.8	50.5
6	39.2	57.6	42.5	23.5	42.8	51.1
7	39.3	57.6	42.4	22.9	43.2	51.3
8	39.3	57.5	42.8	24.3	43.3	51.3
9	38.9	56.5	42.0	22.9	42.4	50.3

表6: 在MS COCO minival集上使用固定尺度8*S*时不同锚框长宽比的分析 (%)。

Aspect Ratio	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
1:4	39.1	57.2	42.3	23.1	43.1	51.4
1:2	39.0	56.9	42.5	23.3	43.5	50.6
1:1	39.3	57.5	42.8	24.3	43.3	51.3
2:1	39.3	57.4	42.3	22.8	43.4	51.0
4:1	39.1	56.9	42.6	22.9	42.9	50.7

锚框尺寸。所提出的方法依赖锚框来定义正样本，我们也研究了锚框尺寸的影响。在之前的实验中，每个位置平铺一个边长为8*S* (*S*的正方形锚框，表示金字塔层级)的总步幅大小为该值。如表5所示，我们使用[5, 6, 7, 8, 9]范围内不同尺度的正方形锚框进行了实验，性能表现相当稳定。此外，如表6所示，我们还对边长为8*S*的锚框进行了不同宽高比的实验。性能对此变化同样不敏感。这些结果表明，所提出的方法对不同锚框设置具有鲁棒性。

4.4. 对比

我们在表8中将最终模型与其他最先进的目标检测器在MS COCO测试开发子集上进行了比较。遵循先前的研究[33, 56]，这些实验采用了多尺度训练策略，*i.e.*，即在训练期间随机选择640到800之间的尺度来调整图像的较短边。此外，我们将总迭代次数加倍至180*K*，并相应地将学习率降低点调整至120*K*和160*K*。其他设置与之前提到的保持一致。

如表8所示，我们采用ResNet-101的方法在未添加任何额外技巧的情况下达到了43.6%的AP，这一结果优于所有使用相同骨干网络的方法，包括Cascade R-CNN [5] (42.8% AP)、C-Mask RCNN [7] (42.0% AP)、RetinaNet [33] (39.1% AP) 和RefineDet [66] (36.4% AP)。通过使用更大的骨干网络ResNeXt-32x8d-101和ResNeXt-64x4d-101 [63]，我们可进一步将所提方法的AP精度分别提升至45.1%和45.6%。其中45.6%的AP结果超越了所有基于锚点和无锚点的检测器，仅比引入改进多尺度训练的SNIP [54] (45.7% AP) 低0.1%。

表7: 在MS COCO minival集上每个位置使用不同数量锚点的结果 (%)。

Method	#sc	#ar	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
RetinaNet (#A=9)	3	3	36.3	55.2	38.8	19.8	39.8	48.8
+Imprs.	3	3	38.4	56.2	41.6	22.2	42.4	50.1
+Imprs.+ATSS	3	3	39.2	57.6	42.7	23.8	42.8	50.9
+Imprs.+ATSS	3	1	39.3	57.7	42.6	23.8	43.5	51.2
+Imprs.+ATSS	1	3	39.2	57.1	42.5	23.2	43.1	50.3
+Imprs.+ATSS	1	1	39.3	57.5	42.8	24.3	43.3	51.3

和测试策略。由于我们的方法涉及正负样本的定义，它与当前大多数技术兼容且互补。我们进一步将可变形卷积网络 (DCN) [10]应用于ResNet和ResNeXt骨干网络以及检测器塔的最后层。DCN持续提升了AP性能：ResNet-101达到46.3%，ResNeXt-32x8d-101达到47.7%，ResNeXt-64x4d-101达到47.7%。最佳结果47.7%通过单模型和单尺度测试实现，大幅超越了所有先前的检测器。最后，采用多尺度测试策略，我们的最佳模型达到了50.7% AP。

4.5. 讨论

先前实验基于RetinaNet，每个位置仅使用一个锚点。基于锚点与无锚点检测器之间仍存在一个尚未探索的差异：每个位置平铺的锚点数量。实际上，原始RetinaNet在每个位置平铺9个锚点 (3种尺度×3种宽高比)，记为RetinaNet (#A=9)，如表7首行所示，其达到36.3% AP。此外，表1中的通用改进也可应用于RetinaNet (#A=9)，将AP性能从36.3%提升至38.4%。在不使用本文提出的ATSS方法时，改进后的RetinaNet (#A=9) 性能优于RetinaNet (#A=1)，即表7中的38.4%对比表1中的37.0%。这些结果表明，在传统的基于IoU的样本选择策略下，每个位置平铺更多锚框是有效的。

然而，在使用我们提出的方法后，会得出相反的结论。具体来说，所提出的ATSS方法也将RetinaNet (#A=9) 的AP提升了0.8%，AP₅₀提升了1.4%，AP₇₅提升了1.1%，达到了与RetinaNet (#A=1) 相近的性能，如表7第三行和第六行所列。此外，当我们将锚框的尺度数量或宽高比数量从3改为1时，结果几乎保持不变，如表7第四行和第五行所示。换言之，只要正样本被适当选择，无论每个位置平铺多少个锚框，结果都是相同的。我们认为，在我们提出的方法下，每个位置平铺多个锚框是一种无用的操作，其正确作用有待进一步研究来揭示。

Table 8: Detection results (%) on MS COCO test-dev set. Bold fonts indicate the best performance.

Method	Data	Backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
<i>anchor-based two-stage:</i>								
MLKP [58]	trainval35	ResNet-101	28.6	52.4	31.6	10.8	33.4	45.1
R-FCN [9]	trainval	ResNet-101	29.9	51.9	-	10.8	32.8	45.0
CoupleNet [74]	trainval	ResNet-101	34.4	54.8	37.2	13.4	38.1	50.8
TDM [53]	trainval	Inception-ResNet-v2-TDM	36.8	57.7	39.2	16.2	39.8	52.1
Hu et al. [18]	trainval35k	ResNet-101	39.0	58.6	42.9	-	-	-
DeepRegionlets [64]	trainval35k	ResNet-101	39.3	59.8	-	21.7	43.7	50.9
FitnessNMS [57]	trainval	DeNet-101	39.5	58.0	42.6	18.9	43.5	54.1
Gu et al. [15]	trainval35k	ResNet-101	39.9	63.1	43.1	22.2	43.4	51.6
DetNet [31]	trainval35k	DetNet-59	40.3	62.1	43.8	23.6	42.6	50.0
Soft-NMS [3]	trainval	ResNet-101	40.8	62.4	44.9	23.0	43.4	53.2
SOD-MTGAN [1]	trainval35k	ResNet-101	41.4	63.2	45.4	24.7	44.2	52.6
G-RMI [19]	trainval35k	Ensemble of Five Models	41.6	61.9	45.4	23.9	43.5	54.9
C-Mask RCNN [7]	trainval35k	ResNet-101	42.0	62.9	46.4	23.4	44.7	53.8
Cascade R-CNN [5]	trainval35k	ResNet-101	42.8	62.1	46.3	23.7	45.5	55.2
Revisiting RCNN [8]	trainval35k	ResNet-101+ResNet-152	43.1	66.1	47.3	25.8	45.9	55.3
SNIP [54]	trainval35k	DPN-98	45.7	67.3	51.1	29.3	48.8	57.1
<i>anchor-based one-stage:</i>								
YOLOv2 [46]	trainval35k	DarkNet-19	21.6	44.0	19.2	5.0	22.4	35.5
SSD512* [36]	trainval35k	VGG-16	28.8	48.5	30.3	10.9	31.8	43.5
STDN513 [69]	trainval	DenseNet-169	31.8	51.0	33.6	14.4	36.1	43.4
DES512 [68]	trainval35k	VGG-16	32.8	53.2	34.5	13.9	36.2	47.5
DSSD513 [12]	trainval35k	ResNet-101	33.2	53.3	35.2	13.0	35.4	51.1
RFB512-E [35]	trainval35k	VGG-16	34.4	55.7	36.4	17.6	37.0	47.6
PPFNet-R512 [21]	trainval35k	VGG-16	35.2	57.6	37.9	18.7	38.6	45.9
RefineDet512 [66]	trainval35k	ResNet-101	36.4	57.5	39.5	16.6	39.9	51.4
RetinaNet [33]	trainval35k	ResNet-101	39.1	59.1	42.3	21.8	42.7	50.2
<i>anchor-free keypoint-based:</i>								
ExtremeNet [71]	trainval35k	Hourglass-104	40.2	55.5	43.2	20.4	43.2	53.1
CornerNet [26]	trainval35k	Hourglass-104	40.5	56.5	43.1	19.4	42.7	53.9
CenterNet-HG [70]	trainval35k	Hourglass-104	42.1	61.1	45.9	24.1	45.5	52.8
Grid R-CNN [39]	trainval35k	ResNeXt-101	43.2	63.0	46.6	25.1	46.5	55.2
CornerNet-Lite [27]	trainval35k	Hourglass-54	43.2	-	-	24.4	44.6	57.3
CenterNet [11]	trainval35k	Hourglass-104	44.9	62.4	48.1	25.6	47.4	57.4
RepPoints [65]	trainval35k	ResNet-101-DCN	45.0	66.1	49.0	26.6	48.6	57.5
<i>anchor-free center-based:</i>								
GA-RPN [59]	trainval35k	ResNet-50	39.8	59.2	43.5	21.8	42.6	50.7
FoveaBox [23]	trainval35k	ResNeXt-101	42.1	61.9	45.2	24.9	46.8	55.6
FSAF [72]	trainval35k	ResNeXt-64x4d-101	42.9	63.8	46.3	26.6	46.2	52.7
FCOS [56]	trainval35k	ResNeXt-64x4d-101	43.2	62.8	46.6	26.5	46.2	53.3
<i>Ours:</i>								
ATSS	trainval35k	ResNet-101	43.6	62.1	47.4	26.1	47.0	53.6
ATSS	trainval35k	ResNeXt-32x8d-101	45.1	63.9	49.1	27.9	48.2	54.6
ATSS	trainval35k	ResNeXt-64x4d-101	45.6	64.6	49.7	28.5	48.9	55.6
ATSS	trainval35k	ResNet-101-DCN	46.3	64.7	50.4	27.7	49.8	58.4
ATSS	trainval35k	ResNeXt-32x8d-101-DCN	47.7	66.6	52.1	29.3	50.8	59.7
ATSS	trainval35k	ResNeXt-64x4d-101-DCN	47.7	66.5	51.9	29.7	50.8	59.4
ATSS (Multi-scale testing)	trainval35k	ResNeXt-32x8d-101-DCN	50.6	68.6	56.1	33.6	52.9	62.2
ATSS (Multi-scale testing)	trainval35k	ResNeXt-64x4d-101-DCN	50.7	68.9	56.3	33.2	52.9	62.4

5. Conclusion

In this work, we point out that the essential difference between one-stage anchor-based and center-based anchor-free detectors is actually the definition of positive and negative training samples. It indicates that how to select positive and negative samples during object detection training is critical. Inspired by that, we delve into this basic issue and propose the adaptive training sample selection, which automatically divides positive and negative training samples according to statistical characteristics of object, hence bridging the gap between anchor-based and anchor-free detectors. We also

discuss the necessity of tiling multiple anchors per location and show that it may not be a so useful operation under current situations. Extensive experiments on the challenging benchmarks MS COCO illustrate that the proposed method can achieve state-of-the-art performances without introducing any additional overhead.

Acknowledgments

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表8: 在MS COCO测试开发集上的检测结果 (%)。粗体表示最佳性能。

Method	Data	Backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
<i>anchor-based two-stage:</i>								
MLKP [58]	trainval35	ResNet-101	28.6	52.4	31.6	10.8	33.4	45.1
R-FCN [9]	trainval	ResNet-101	29.9	51.9	-	10.8	32.8	45.0
CoupleNet [74]	trainval	ResNet-101	34.4	54.8	37.2	13.4	38.1	50.8
TDM [53]	trainval	Inception-ResNet-v2-TDM	36.8	57.7	39.2	16.2	39.8	52.1
Hu et al. [18]	trainval35k	ResNet-101	39.0	58.6	42.9	-	-	-
DeepRegionlets [64]	trainval35k	ResNet-101	39.3	59.8	-	21.7	43.7	50.9
FitnessNMS [57]	trainval	DeNet-101	39.5	58.0	42.6	18.9	43.5	54.1
Gu et al. [15]	trainval35k	ResNet-101	39.9	63.1	43.1	22.2	43.4	51.6
DetNet [31]	trainval35k	DetNet-59	40.3	62.1	43.8	23.6	42.6	50.0
Soft-NMS [3]	trainval	ResNet-101	40.8	62.4	44.9	23.0	43.4	53.2
SOD-MTGAN [1]	trainval35k	ResNet-101	41.4	63.2	45.4	24.7	44.2	52.6
G-RMI [19]	trainval35k	Ensemble of Five Models	41.6	61.9	45.4	23.9	43.5	54.9
C-Mask RCNN [7]	trainval35k	ResNet-101	42.0	62.9	46.4	23.4	44.7	53.8
Cascade R-CNN [5]	trainval35k	ResNet-101	42.8	62.1	46.3	23.7	45.5	55.2
Revisiting RCNN [8]	trainval35k	ResNet-101+ResNet-152	43.1	66.1	47.3	25.8	45.9	55.3
SNIP [54]	trainval35k	DPN-98	45.7	67.3	51.1	29.3	48.8	57.1
<i>anchor-based one-stage:</i>								
YOLOv2 [46]	trainval35k	DarkNet-19	21.6	44.0	19.2	5.0	22.4	35.5
SSD512* [36]	trainval35k	VGG-16	28.8	48.5	30.3	10.9	31.8	43.5
STDN513 [69]	trainval	DenseNet-169	31.8	51.0	33.6	14.4	36.1	43.4
DES512 [68]	trainval35k	VGG-16	32.8	53.2	34.5	13.9	36.2	47.5
DSSD513 [12]	trainval35k	ResNet-101	33.2	53.3	35.2	13.0	35.4	51.1
RFB512-E [35]	trainval35k	VGG-16	34.4	55.7	36.4	17.6	37.0	47.6
PFPNet-R512 [21]	trainval35k	VGG-16	35.2	57.6	37.9	18.7	38.6	45.9
RefineDet512 [66]	trainval35k	ResNet-101	36.4	57.5	39.5	16.6	39.9	51.4
RetinaNet [33]	trainval35k	ResNet-101	39.1	59.1	42.3	21.8	42.7	50.2
<i>anchor-free keypoint-based:</i>								
ExtremeNet [71]	trainval35k	Hourglass-104	40.2	55.5	43.2	20.4	43.2	53.1
CornerNet [26]	trainval35k	Hourglass-104	40.5	56.5	43.1	19.4	42.7	53.9
CenterNet-HG [70]	trainval35k	Hourglass-104	42.1	61.1	45.9	24.1	45.5	52.8
Grid R-CNN [39]	trainval35k	ResNeXt-101	43.2	63.0	46.6	25.1	46.5	55.2
CornerNet-Lite [27]	trainval35k	Hourglass-54	43.2	-	-	24.4	44.6	57.3
CenterNet [11]	trainval35k	Hourglass-104	44.9	62.4	48.1	25.6	47.4	57.4
RepPoints [65]	trainval35k	ResNet-101-DCN	45.0	66.1	49.0	26.6	48.6	57.5
<i>anchor-free center-based:</i>								
GA-RPN [59]	trainval35k	ResNet-50	39.8	59.2	43.5	21.8	42.6	50.7
FoveaBox [23]	trainval35k	ResNeXt-101	42.1	61.9	45.2	24.9	46.8	55.6
FSAF [72]	trainval35k	ResNeXt-64x4d-101	42.9	63.8	46.3	26.6	46.2	52.7
FCOS [56]	trainval35k	ResNeXt-64x4d-101	43.2	62.8	46.6	26.5	46.2	53.3
<i>Ours:</i>								
ATSS	trainval35k	ResNet-101	43.6	62.1	47.4	26.1	47.0	53.6
ATSS	trainval35k	ResNeXt-32x8d-101	45.1	63.9	49.1	27.9	48.2	54.6
ATSS	trainval35k	ResNeXt-64x4d-101	45.6	64.6	49.7	28.5	48.9	55.6
ATSS	trainval35k	ResNet-101-DCN	46.3	64.7	50.4	27.7	49.8	58.4
ATSS	trainval35k	ResNeXt-32x8d-101-DCN	47.7	66.6	52.1	29.3	50.8	59.7
ATSS	trainval35k	ResNeXt-64x4d-101-DCN	47.7	66.5	51.9	29.7	50.8	59.4
ATSS (Multi-scale testing)	trainval35k	ResNeXt-32x8d-101-DCN	50.6	68.6	56.1	33.6	52.9	62.2
ATSS (Multi-scale testing)	trainval35k	ResNeXt-64x4d-101-DCN	50.7	68.9	56.3	33.2	52.9	62.4

5. 结论

在这项工作中, 我们指出单阶段基于锚框的检测器与基于中心的免锚框检测器之间的本质差异, 实际上在于正负训练样本的定义方式。这表明在目标检测训练过程中如何选择正负样本至关重要。受此启发, 我们深入探究这一根本问题, 提出了自适应训练样本选择方法, 该方法能根据目标的统计特征自动划分正负训练样本, 从而弥合了基于锚框与免锚框检测器之间的鸿沟。我们还

讨论每个位置平铺多个锚点的必要性, 并表明在当前情况下这可能不是一个非常有用的操作。在具有挑战性的基准测试MS COCO上进行的大量实验表明, 所提出的方法可以在不引入任何额外开销的情况下实现最先进的性能。

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References

- [1] Yancheng Bai, Yongqiang Zhang, Mingli Ding, and Bernard Ghanem. SOD-MTGAN: small object detection via multi-task generative adversarial network. In *ECCV*, 2018. 8
- [2] Sean Bell, C. Lawrence Zitnick, Kavita Bala, and Ross B. Girshick. Inside-outside net: Detecting objects in context with skip pooling and recurrent neural networks. In *CVPR*, 2016. 2
- [3] Navaneeth Bodla, Bharat Singh, Rama Chellappa, and Larry S. Davis. Improving object detection with one line of code. In *ICCV*, 2017. 8
- [4] Zhaowei Cai, Quanfu Fan, Rogério Schmidt Feris, and Nuno Vasconcelos. A unified multi-scale deep convolutional neural network for fast object detection. In *ECCV*, 2016. 2
- [5] Zhaowei Cai and Nuno Vasconcelos. Cascade R-CNN: delving into high quality object detection. In *CVPR*, 2018. 2, 7, 8
- [6] Kean Chen, Jianguo Li, Weiyao Lin, John See, Ji Wang, Lingyu Duan, Zhibo Chen, Changwei He, and Junni Zou. Towards accurate one-stage object detection with ap-loss. In *CVPR*, 2019. 2
- [7] Zhe Chen, Shaoli Huang, and Dacheng Tao. Context refinement for object detection. In *ECCV*, 2018. 2, 7, 8
- [8] Bowen Cheng, Yunchao Wei, Honghui Shi, Rogério Schmidt Feris, Jinjun Xiong, and Thomas S. Huang. Revisiting RCNN: on awakening the classification power of faster RCNN. In *ECCV*, 2018. 8
- [9] Jifeng Dai, Yi Li, Kaiming He, and Jian Sun. R-FCN: object detection via region-based fully convolutional networks. In *NIPS*, 2016. 1, 2, 8
- [10] Jifeng Dai, Haozhi Qi, Yuwen Xiong, Yi Li, Guodong Zhang, Han Hu, and Yichen Wei. Deformable convolutional networks. In *ICCV*, 2017. 7
- [11] Kaiwen Duan, Song Bai, Lingxi Xie, Honggang Qi, Qingming Huang, and Qi Tian. Centernet: Keypoint triplets for object detection. In *ICCV*, 2019. 2, 8
- [12] Cheng-Yang Fu, Wei Liu, Ananth Ranga, Amrith Tyagi, and Alexander C. Berg. DSSD : Deconvolutional single shot detector. *CoRR*, 2017. 2, 8
- [13] Ross B. Girshick. Fast R-CNN. In *ICCV*, 2015. 2
- [14] Ross B. Girshick, Jeff Donahue, Trevor Darrell, and Jitendra Malik. Rich feature hierarchies for accurate object detection and semantic segmentation. In *CVPR*, 2014. 2
- [15] Jiayuan Gu, Han Hu, Liwei Wang, Yichen Wei, and Jifeng Dai. Learning region features for object detection. In *ECCV*, 2018. 8
- [16] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016. 3
- [17] Yihui He, Chenchen Zhu, Jianren Wang, Marios Savvides, and Xiangyu Zhang. Bounding box regression with uncertainty for accurate object detection. In *CVPR*, 2019. 2
- [18] Han Hu, Jiayuan Gu, Zheng Zhang, Jifeng Dai, and Yichen Wei. Relation networks for object detection. In *CVPR*, 2018. 8
- [19] Jonathan Huang, Vivek Rathod, Chen Sun, Menglong Zhu, Anoop Korattikara, Alireza Fathi, Ian Fischer, Zbigniew Wojna, Yang Song, Sergio Guadarrama, and Kevin Murphy. Speed/accuracy trade-offs for modern convolutional object detectors. In *CVPR*, 2017. 8
- [20] Lichao Huang, Yi Yang, Yafeng Deng, and Yinan Yu. Densebox: Unifying landmark localization with end to end object detection. *CoRR*, 2015. 3
- [21] Seung-Wook Kim, Hyong-Keun Kook, Jee-Young Sun, Mun-Cheon Kang, and Sung-Jea Ko. Parallel feature pyramid network for object detection. In *ECCV*, 2018. 2, 8
- [22] Tao Kong, Fuchun Sun, Wen-bing Huang, and Huaping Liu. Deep feature pyramid reconfiguration for object detection. In *ECCV*, 2018. 2
- [23] Tao Kong, Fuchun Sun, Huaping Liu, Yuning Jiang, and Jianbo Shi. Foveabox: Beyond anchor-based object detector. *CoRR*, 2019. 1, 3, 8
- [24] Tao Kong, Fuchun Sun, Anbang Yao, Huaping Liu, Ming Lu, and Yurong Chen. RON: reverse connection with objectness prior networks for object detection. In *CVPR*, 2017. 2
- [25] Tao Kong, Anbang Yao, Yurong Chen, and Fuchun Sun. Hypernet: Towards accurate region proposal generation and joint object detection. In *CVPR*, 2016. 2
- [26] Hei Law and Jia Deng. Cornernet: Detecting objects as paired keypoints. In *ECCV*, 2018. 1, 2, 8
- [27] Hei Law, Yun Teng, Olga Russakovsky, and Jia Deng. Cornernet-lite: Efficient keypoint based object detection. *CoRR*, 2019. 2, 8
- [28] Hyungtae Lee, Sungmin Eum, and Heesung Kwon. ME R-CNN: multi-expert region-based CNN for object detection. In *ICCV*, 2017. 2
- [29] Shuai Li, Lingxiao Yang, Jianqiang Huang, Xian-Sheng Hua, and Lei Zhang. Dynamic anchor feature selection for single-shot object detection. In *ICCV*, 2019. 2
- [30] Yanghao Li, Yuntao Chen, Naiyan Wang, and Zhaoxiang Zhang. Scale-aware trident networks for object detection. In *ICCV*, 2019. 2
- [31] Zeming Li, Chao Peng, Gang Yu, Xiangyu Zhang, Yangdong Deng, and Jian Sun. Detnet: A backbone network for object detection. In *ECCV*, 2018. 8
- [32] Tsung-Yi Lin, Piotr Dollár, Ross B. Girshick, Kaiming He, Bharath Hariharan, and Serge J. Belongie. Feature pyramid networks for object detection. In *CVPR*, 2017. 1, 2
- [33] Tsung-Yi Lin, Priya Goyal, Ross B. Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *ICCV*, 2017. 1, 2, 3, 7, 8
- [34] Tsung-Yi Lin, Michael Maire, Serge J. Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: common objects in context. In *ECCV*, 2014. 2, 3
- [35] Songtao Liu, Di Huang, and Yunhong Wang. Receptive field block net for accurate and fast object detection. In *ECCV*, 2018. 2, 8
- [36] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott E. Reed, Cheng-Yang Fu, and Alexander C. Berg. SSD: single shot multibox detector. In *ECCV*, 2016. 1, 2, 8

参考文献

[1] 白岩城、张永强、丁明利、Bernard Ghanem. SOD-MTG AN: 通过多任务生成对抗网络进行小目标检测。发表于 *ECCV*, 2018年。8[2] Sean Bell、C. Lawrence Zitnick、Kavita Bala、Ross B. Girshick. 内外网: 通过跳跃池化和循环神经网络在上下文中检测目标。发表于 *CVPR*, 2016年。2[3] Navaneeth Bodla、Bharat Singh、Rama Chellappa、Larry S. Davis。用一行代码改进目标检测。发表于 *ICCV*, 2017年。8[4] 蔡兆伟、范全富、Rogério Schmidt Feris、Nuno Vasconcelos。用于快速目标检测的统一多尺度深度卷积神经网络。发表于 *ECCV*, 2016年。2[5] 蔡兆伟、Nuno Vasconcelos。Cascade R-CNN: 深入高质量目标检测。发表于 *CVPR*, 2018年。2, 7, 8[6] 陈侃、李建国、林伟尧、John See、王吉、段凌宇、陈志波、何昌伟、邹俊妮。通过AP损失实现精确的单阶段目标检测。发表于 *CVPR*, 2019年。2[7] 陈哲、黄少立、陶大程。用于目标检测的上下文细化。发表于 *ECCV*, 2018年。2, 7, 8[8] 程博文、魏云超、施鸿辉、Rogério Schmidt Feris、熊金俊、黄煦涛。重新审视RCNN: 唤醒Faster RCNN的分类能力。发表于 *ECCV*, 2018年。8[9] 代季峰、李轶、何恺明、孙剑。R-FCN: 通过基于区域的全卷积网络进行目标检测。发表于 *NIPS*, 2016年。1, 2, 8[10] 代季峰、齐浩志、熊宇雯、李轶、张国栋、胡翰、韦毅辰。可变形卷积网络。发表于 *ICCV*, 2017年。7[11] 段凯文、白雪、谢凌曦、齐洪刚、黄庆明、田奇。Centernet: 用于目标检测的关键点三元组。发表于 *ICCV*, 2019年。2, 8[12] Cheng-Yang Fu、Wei Liu、Ananth Ranga、Amrith Tyagi、Alexander C. Berg。DSSD: 反卷积单次检测器。 *CoRR*, 2017年。2, 8[13] Ross B. Girshick。Fast R-CNN。发表于 *ICCV*, 2015年。2[14] Ross B. Girshick、Jeff Donahue、Trevor Darrell、Jitendra Malik。用于精确目标检测和语义分割的丰富特征层次结构。发表于 *CVPR*, 2014年。2[15] 顾家园、胡翰、王立威、韦毅辰、代季峰。学习用于目标检测的区域特征。发表于 *ECCV*, 2018年。8[16] 何恺明、张祥雨、任少卿、孙剑。用于图像识别的深度残差学习。发表于 *CVPR*, 2016年。3[17] 何一灰、朱晨辰、王建人、Marios Savvides、张祥雨。用于精确目标检测的带不确定性的边界框回归。发表于 *CVPR*, 2019年。2[18] 胡翰、顾家园、张政、代季峰、韦毅辰。用于目标检测的关系网络。发表于 *CVPR*, 2018年。8

[19] Jonathan Huang、Vivek Rathod、Chen Sun、Menglong Zhu、Anoop Korattikara、Alireza Fathi、Ian Fischer、Zbigniew Wojna、Yang Song、Sergio Guadarrama、Kevin Murphy。现代卷积目标检测器的速度/精度权衡。载于 *CVPR*, 2017年。8 [20] Lichao Huang、Yi Yang、Yafeng Deng、Yinan Yu。DenseBox: 将地标定位与端到端目标检测统一。 *CoRR*, 2015年。3 [21] Seung-Wook Kim、Hyong-Keun Kook、Jee-Young Sun、Mun-Cheon Kang、Sung-Jea Ko。用于目标检测的并行特征金字塔网络。载于 *ECCV*, 2018年。2, 8 [22] Tao Kong、Fuchun Sun、Wen-bing Huang、Huaping Liu。用于目标检测的深度特征金字塔重构。载于 *ECCV*, 2018年。2 [23] Tao Kong、Fuchun Sun、Huaping Liu、Yuning Jiang、Jianbo Shi。FoveaBox: 超越基于锚框的目标检测器。 *CoRR*, 2019年。1, 3, 8 [24] Tao Kong、Fuchun Sun、Anbang Yao、Huaping Liu、Ming Lu、Yurong Chen。RON: 结合目标性先验的反向连接网络用于目标检测。载于 *CVPR*, 2017年。2 [25] Tao Kong、Anbang Yao、Yurong Chen、Fuchun Sun。HyperNet: 面向精确区域提议生成与联合目标检测。载于 *CVPR*, 2016年。2 [26] Hei Law、Jia Deng。CornerNet: 将目标检测为成对关键点。载于 *ECCV*, 2018年。1, 2, 8 [27] Hei Law、Yun Teng、Olga Russakovsky、Jia Deng。CornerNet-Lite: 高效的基于关键点目标检测。 *CoRR*, 2019年。2, 8 [28] Hyungtae Lee、Sungmin Eum、Heesung Kwon。ME R-CNN: 用于目标检测的多专家基于区域的CNN。载于 *ICCV*, 2017年。2 [29] Shuai Li、Lingxiao Yang、Jianqiang Huang、Xian-Sheng Hua、Lei Zhang。用于单次目标检测的动态锚框特征选择。载于 *ICCV*, 2019年。2 [30] Yanghao Li、Yuntao Chen、Naiyan Wang、Zhaoxiang Zhang。用于目标检测的尺度感知三叉戟网络。载于 *ICCV*, 2019年。2 [31] Zeming Li、Chao Peng、Gang Yu、Xiangyu Zhang、Yangdong Deng、Jian Sun。DetNet: 一种用于目标检测的骨干网络。载于 *ECCV*, 2018年。8 [32] Tsung-Yi Lin、Piotr Dollár、Ross B. Girshick、Kaiming He、Bharath Hariharan、Serge J. Belongie。用于目标检测的特征金字塔网络。载于 *CVPR*, 2017年。1, 2 [33] Tsung-Yi Lin、Priya Goyal、Ross B. Girshick、Kaiming He、Piotr Dollár。用于密集目标检测的焦点损失。载于 *ICCV*, 2017年。1, 2, 3, 7, 8 [34] Tsung-Yi Lin、Michael Maire、Serge J. Belongie、James Hays、Pietro Perona、Deva Ramanan、Piotr Dollár、C. Lawrence Zitnick。Microsoft COCO: 上下文中的常见目标。载于 *ECCV*, 2014年。2, 3 [35] Songtao Liu、Di Huang、Yunhong Wang。用于精确快速目标检测的感受野块网络。载于 *ECCV*, 2018年。2, 8 [36] Wei Liu、Dragomir Anguelov、Dumitru Erhan、Christian Szegedy、Scott E. Reed、Cheng-Yang Fu、Alexander C. Berg。SSD: 单次多框检测器。载于 *ECCV*, 2016年。1, 2, 8

- [37] Wei Liu, Shengcai Liao, Weiqiang Ren, Weidong Hu, and Yinan Yu. High-level semantic feature detection: A new perspective for pedestrian detection. In *CVPR*, 2019. 3
- [38] Yong Liu, Ruiping Wang, Shiguang Shan, and Xilin Chen. Structure inference net: Object detection using scene-level context and instance-level relationships. In *CVPR*, 2018. 2
- [39] Xin Lu, Buyu Li, Yuxin Yue, Quanquan Li, and Junjie Yan. Grid R-CNN. In *CVPR*, 2019. 2, 8
- [40] Mahyar Najibi, Mohammad Rastegari, and Larry S. Davis. G-CNN: an iterative grid based object detector. In *CVPR*, 2016. 2
- [41] Mahyar Najibi, Bharat Singh, and Larry S. Davis. Autofocus: Efficient multi-scale inference. In *ICCV*, 2019. 2
- [42] Jing Nie, Rao Muhammad Anwer, Hisham Cholakkal, Fahad Shahbaz Khan, Yanwei Pang, and Ling Shao. Enriched feature guided refinement network for object detection. In *ICCV*, 2019. 2
- [43] Jiangmiao Pang, Kai Chen, Jianping Shi, Huajun Feng, Wanli Ouyang, and Dahua Lin. Libra R-CNN: towards balanced learning for object detection. In *CVPR*, 2019. 2
- [44] Zheng Qin, Zeming Li, Zhaoning Zhang, Yiping Bao, Gang Yu, Yuxing Peng, and Jian Sun. Thundernet: Towards real-time generic object detection. In *ICCV*, 2019. 2
- [45] Joseph Redmon, Santosh Kumar Divvala, Ross B. Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *CVPR*, 2016. 2
- [46] Joseph Redmon and Ali Farhadi. YOLO9000: better, faster, stronger. In *CVPR*, 2017. 8
- [47] Shaoqing Ren, Kaiming He, Ross B. Girshick, and Jian Sun. Faster R-CNN: towards real-time object detection with region proposal networks. *TPAMI*, 2017. 1, 2
- [48] Hamid Rezatofighi, Nathan Tsoi, JunYoung Gwak, Amir Sadeghian, Ian D. Reid, and Silvio Savarese. Generalized intersection over union: A metric and a loss for bounding box regression. In *CVPR*, 2019. 3
- [49] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael S. Bernstein, Alexander C. Berg, and Fei-Fei Li. Imagenet large scale visual recognition challenge. *IJCV*, 2015. 3
- [50] Zhiqiang Shen, Zhuang Liu, Jianguo Li, Yu-Gang Jiang, Yurong Chen, and Xiangyang Xue. DSOD: learning deeply supervised object detectors from scratch. In *ICCV*, 2017. 2
- [51] Abhinav Shrivastava and Abhinav Gupta. Contextual priming and feedback for faster R-CNN. In *ECCV*, 2016. 2
- [52] Abhinav Shrivastava, Abhinav Gupta, and Ross B. Girshick. Training region-based object detectors with online hard example mining. In *CVPR*, 2016. 2
- [53] Abhinav Shrivastava, Rahul Sukthankar, Jitendra Malik, and Abhinav Gupta. Beyond skip connections: Top-down modulation for object detection. *CoRR*, 2016. 8
- [54] Bharat Singh and Larry S. Davis. An analysis of scale invariance in object detection - SNIP. In *CVPR*, 2018. 2, 7, 8
- [55] Zhiyu Tan, Xuecheng Nie, Qi Qian, Nan Li, and Hao Li. Learning to rank proposals for object detection. In *ICCV*, 2019. 2
- [56] Zhi Tian, Chunhua Shen, Hao Chen, and Tong He. FCOS: fully convolutional one-stage object detection. In *ICCV*, 2019. 1, 2, 3, 7, 8
- [57] Lachlan Tychsen-Smith and Lars Petersson. Improving object localization with fitness NMS and bounded iou loss. In *CVPR*, 2018. 8
- [58] Hao Wang, Qilong Wang, Mingqi Gao, Peihua Li, and Wangmeng Zuo. Multi-scale location-aware kernel representation for object detection. In *CVPR*, 2018. 8
- [59] Jiaqi Wang, Kai Chen, Shuo Yang, Chen Change Loy, and Dahua Lin. Region proposal by guided anchoring. In *CVPR*, 2019. 3, 8
- [60] Tiancai Wang, Rao Muhammad Anwer, Hisham Cholakkal, Fahad Shahbaz Khan, Yanwei Pang, and Ling Shao. Learning rich features at high-speed for single-shot object detection. In *ICCV*, 2019. 2
- [61] Xiaolong Wang, Abhinav Shrivastava, and Abhinav Gupta. A-fast-rcnn: Hard positive generation via adversary for object detection. In *CVPR*, 2017. 2
- [62] Yuxin Wu and Kaiming He. Group normalization. In *ECCV*, 2018. 3
- [63] Saining Xie, Ross B. Girshick, Piotr Dollár, Zhuowen Tu, and Kaiming He. Aggregated residual transformations for deep neural networks. In *CVPR*, 2017. 7
- [64] Hongyu Xu, Xutao Lv, Xiaoyu Wang, Zhou Ren, Navaneeth Bodla, and Rama Chellappa. Deep regionlets for object detection. In *ECCV*, 2018. 8
- [65] Ze Yang, Shaohui Liu, Han Hu, Liwei Wang, and Stephen Lin. Reppoints: Point set representation for object detection. In *ICCV*, 2019. 2, 8
- [66] Shifeng Zhang, Longyin Wen, Xiao Bian, Zhen Lei, and Stan Z. Li. Single-shot refinement neural network for object detection. In *CVPR*, 2018. 2, 7, 8
- [67] Xiaosong Zhang, Fang Wan, Chang Liu, Rongrong Ji, and Qixiang Ye. Freeanchor: Learning to match anchors for visual object detection. In *NIPS*, 2019. 2
- [68] Zhishuai Zhang, Siyuan Qiao, Cihang Xie, Wei Shen, Bo Wang, and Alan L. Yuille. Single-shot object detection with enriched semantics. In *CVPR*, 2018. 2, 8
- [69] Peng Zhou, Bingbing Ni, Cong Geng, Jianguo Hu, and Yi Xu. Scale-transferrable object detection. In *CVPR*, 2018. 2, 8
- [70] Xingyi Zhou, Dequan Wang, and Philipp Krähenbühl. Objects as points. *CoRR*, 2019. 2, 8
- [71] Xingyi Zhou, Jiacheng Zhuo, and Philipp Krähenbühl. Bottom-up object detection by grouping extreme and center points. In *CVPR*, 2019. 1, 2, 8
- [72] Chenchen Zhu, Yihui He, and Marios Savvides. Feature selective anchor-free module for single-shot object detection. In *CVPR*, 2019. 3, 8
- [73] Rui Zhu, Shifeng Zhang, Xiaobo Wang, Longyin Wen, Hailin Shi, Liefeng Bo, and Tao Mei. Scratchdet: Exploring to train single-shot object detectors from scratch. *CoRR*, 2018. 2
- [74] Yousong Zhu, Chaoyang Zhao, Jinqiao Wang, Xu Zhao, Yi Wu, and Hanqing Lu. Couplet: Coupling global structure with local parts for object detection. In *ICCV*, 2017. 8

[37] 刘伟、廖圣才、任伟强、胡卫东、于毅南。高层语义特征检测：行人检测的新视角。发表于*CVPR*，2019年。3[38] 刘勇、王瑞平、山世光、陈熙霖。结构推理网络：利用场景级上下文和实例级关系的目标检测。发表于*CVPR*，2018年。2[39] 卢欣、李步宇、岳雨昕、李全全、颜俊杰。Grid R-CNN。发表于*CVPR*，2019年。2, 8[40] Mahyar Najibi, Mohammad Rastegari, Larry S. Davis. G-CNN：一种基于迭代网格的目标检测器。发表于*CVPR*，2016年。2[41] Mahyar Najibi, Bharat Singh, Larry S. Davis. Autofocus：高效的多尺度推理。发表于*ICCV*，2019年。2[42] 聂晶、Rao Muhammad Anwer, Hisham Cholakkal, Fahad Shahbaz Khan, 庞彦伟、邵岭。用于目标检测的丰富特征引导精炼网络。发表于*ICCV*，2019年。2[43] 庞江森、陈恺、石建萍、冯华君、欧阳万里、林达华。Libra R-CNN：迈向目标检测的平衡学习。发表于*CVPR*，2019年。2[44] 覃政、李泽明、张昭宁、鲍一平、余刚、彭宇行、孙剑。ThunderNet：迈向实时通用目标检测。发表于*ICCV*，2019年。2[45] Joseph Redmon, Santosh Kumar Divvala, Ross B. Girshick, Ali Farhadi. You Only Look Once：统一的实时目标检测。发表于*CVPR*，2016年。2[46] Joseph Redmon, Ali Farhadi. YOLO9000：更好、更快、更强。发表于*CVPR*，2017年。8[47] 任少卿、何恺明、Ross B. Girshick, 孙剑。Faster R-CNN：利用区域提议网络实现实时目标检测。*TPAMI*，2017年。1, 2[48] Hamid Rezaatofghi, Nathan Tsoi, JunYoung Gwak, Amir Sadeghian, Ian D. Reid, Silvio Savarese. 广义交并比：一种边界框回归的度量与损失函数。发表于*CVPR*，2019年。3[49] Olga Russakovsky, 贾佳、苏昊、Jonathan Krause、Sanjeev Satheesh、马帅、黄志恒、Andrej Karpathy, Aditya Khosla, Michael S. Bernstein, Alexander C. Berg, 李飞飞。ImageNet大规模视觉识别挑战赛。*IJCV*，2015年。3[50] 沈志强、刘壮、李建国、姜育刚、陈玉荣、薛向阳。DSOD：从零开始学习深度监督的目标检测器。发表于*ICCV*，2017年。2[51] Abhinav Shrivastava, Abhinav Gupta。用于Faster R-CNN的上下文启动与反馈。发表于*ECCV*，2016年。2[52] Abhinav Shrivastava, Abhinav Gupta, Ross B. Girshick。使用在线难例挖掘训练基于区域的目标检测器。发表于*CVPR*，2016年。2[53] Abhinav Shrivastava, Rahul Sukthankar, Jitendra Malik, Abhinav Gupta。超越跳跃连接：用于目标检测的自顶向下调制。*CoRR*，2016年。8[54] Bharat Singh, Larry S. Davis。目标检测中尺度不变性分析 - SNIP。发表于*CVPR*，2018年。2, 7, 8[55] 谭智宇、聂雪成、钱琦、李楠、李浩。学习对目标检测的提议进行排序。发表于*ICCV*，2019年。2

[56] 田植、沈春华、陈浩、何通。FCOS：全卷积单阶段目标检测。发表于*ICCV*，2019年。1, 2, 3, 7, 8[57] Lachlan Tychsen-Smith与Lars Petersson。通过适应度NMS和有界IoU损失改进目标定位。发表于*CVPR*，2018年。8[58] 王浩、王启龙、高铭琪、李培华、左旺孟。用于目标检测的多尺度位置感知核表示。发表于*CVPR*，2018年。8[59] 王佳祺、陈凯、杨硕、罗诚章、林达华。基于引导锚定的区域提议。发表于*CVPR*，2019年。3, 8[60] 王天财、Rao Muhammad Anwer, Hisham Cholakkal, Fahad Shahbaz Khan, 庞彦伟、邵岭。面向单发目标检测的高速丰富特征学习。发表于*ICCV*，2019年。2[61] 王晓龙、Abhinav Shrivastava, Abhinav Gupta. A-Fast-RCNN：通过对抗生成困难正样本用于目标检测。发表于*CVPR*，2017年。2[62] 吴育昕、何恺明。组归一化。发表于*ECCV*，2018年。3[63] 谢赛宁、Ross B. Girshick, Piotr Dollár, 涂卓文、何恺明。深度神经网络的聚合残差变换。发表于*CVPR*，2017年。7[64] 徐宏宇、吕旭涛、王晓宇、任舟、Navaneeth Bodla, Rama Chellappa。用于目标检测的深度区域网络。发表于*ECCV*，2018年。8[65] 杨泽、刘绍辉、胡涵、王立威、林惊。RepPoints：目标检测的点集表示。发表于*ICCV*，2019年。2, 8[66] 张世峰、文龙吟、边潇、雷震、Li Stan Z。用于目标检测的单发优化神经网络。发表于*CVPR*，2018年。2, 7, 8[67] 张晓松、万方、刘畅、纪荣荣、叶齐祥。Free Anchor：学习匹配视觉目标检测的锚点。发表于*NIPS*，2019年。2[68] 张智帅、乔思源、谢慈航、沈玮、王波、Alan L. Yuille。具有丰富语义的单发目标检测。发表于*CVPR*，2018年。2, 8[69] 周鹏、倪冰冰、耿聪、胡建国、徐奕。尺度可迁移的目标检测。发表于*CVPR*，2018年。2, 8[70] 周行易、Dequan Wang, Philipp Krähenbühl。目标即点。*CoRR*，2019年。2, 8[71] 周行易、Jiacheng Zhuo, Philipp Krähenbühl。通过分组极值点和中心点的自底向上目标检测。发表于*CVPR*，2019年。1, 2, 8[72] 朱晨琛、何一辉、Marios Savvides。用于单发目标检测的特征选择无锚模块。发表于*CVPR*，2019年。3, 8[73] 朱锐、张世峰、王潇波、文龙吟、施海林、薄列峰、梅涛。ScratchDet：探索从头训练单发目标检测器。*CoRR*，2018年。2[74] 朱友松、赵朝阳、王金桥、赵旭、吴毅、卢汉卿。CoupleNet：将全局结构与局部部件耦合用于目标检测。发表于*ICCV*，2017年。8